

Digital Design & Computer Arch.

Lecture 6: Timing and Verification

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Readings (This Week)

- Hardware Description Languages and Verilog
 - H&H Chapter 4 in full
- Timing and Verification
 - H&H Chapters 2.9 and 3.5 + (start Chapter 5)
- By this week, make sure you are done with
 - **P&P Chapters 1-3 + H&H Chapters 1-4**

Readings (Next Week)

■ Von Neumann Model, LC-3, and MIPS

- ❑ P&P, Chapter 4, 5
- ❑ H&H, Chapter 6
- ❑ P&P, Appendices A and C (ISA and microarchitecture of LC-3)
- ❑ H&H, Appendix B (MIPS instructions)

■ Programming

- ❑ P&P, Chapter 6

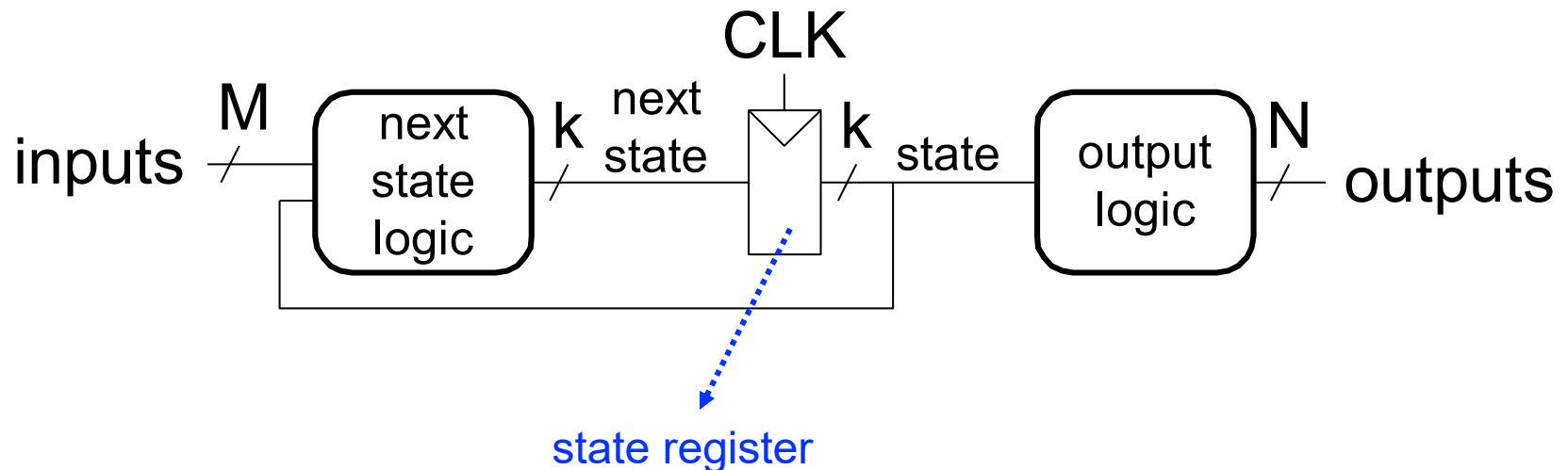
■ **Recommended:** Digital Building Blocks

- ❑ H&H, Chapter 5

Let's Wrap Up HDL and Verilog

Recall: Finite State Machines (FSMs)

- Each FSM consists of three separate parts:
 - next state logic
 - state register
 - output logic

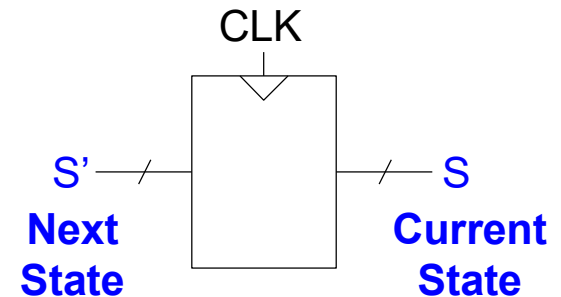


At the beginning of the clock cycle, next state is latched into the state register

Recall: Finite State Machines (FSMs) Consist of:

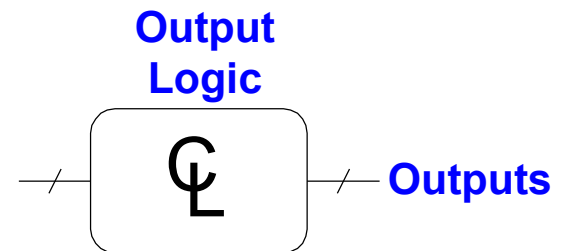
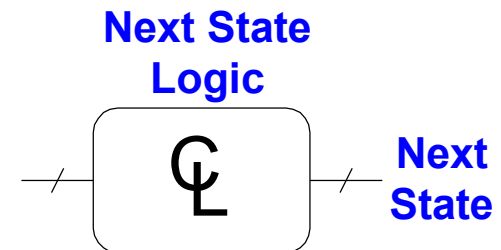
■ Sequential Circuits

- State register(s)
 - Store the current state and
 - Provide the next state at the clock edge



■ Combinational Circuits

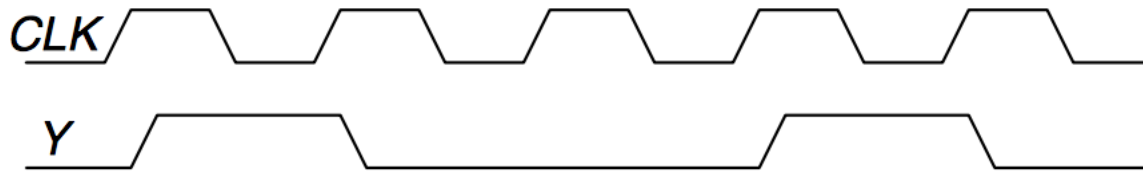
- Next state logic
 - Determines what the next state will be
- Output logic
 - Generates the outputs



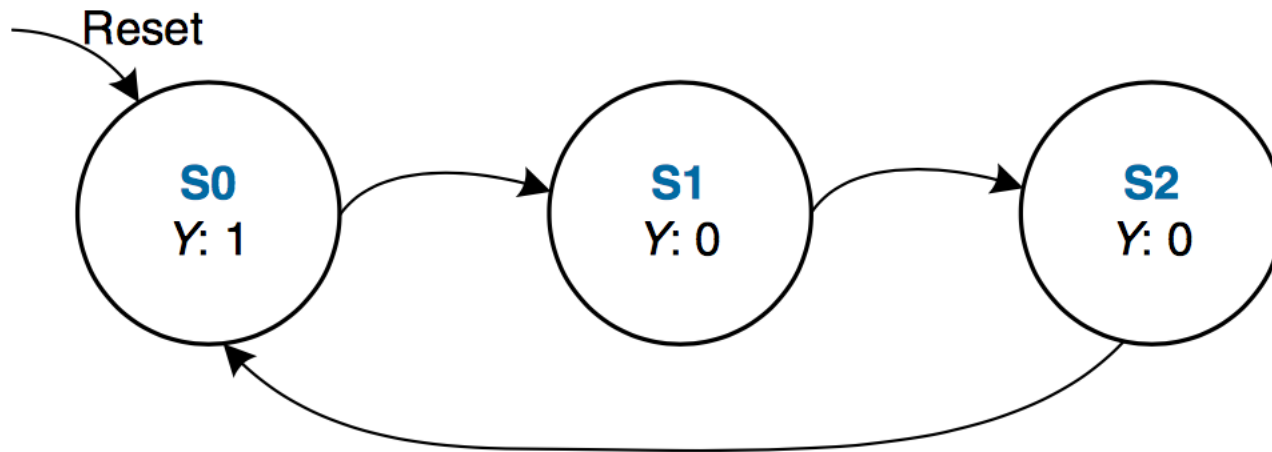
Recall: FSM Design Procedure

- **Determine** all possible states of your machine
- **Develop** a **state transition diagram**
 - Generally this is done from a textual description
 - You need to 1) determine the **inputs** and **outputs** for each **state** and 2) figure out how to get from one state to another
- **Approach**
 - Start by defining the **reset state** and what happens from it – this is typically an easy point to start from
 - Then continue to add **transitions** and **states**
 - Picking **good state names** is very important
 - Building an FSM is **like** programming (but it *is not* programming!)
 - An FSM has a sequential “control-flow” like a program with conditionals and goto’s
 - The if-then-else construct is controlled by one or more inputs
 - The outputs are controlled by the state or the inputs
 - In hardware, we typically have many concurrent FSMs

FSM Example 1: Divide the Clock Frequency by 3



The output Y is HIGH for **one clock cycle out of every 3**. In other words, the output **divides the frequency of the clock by 3**.

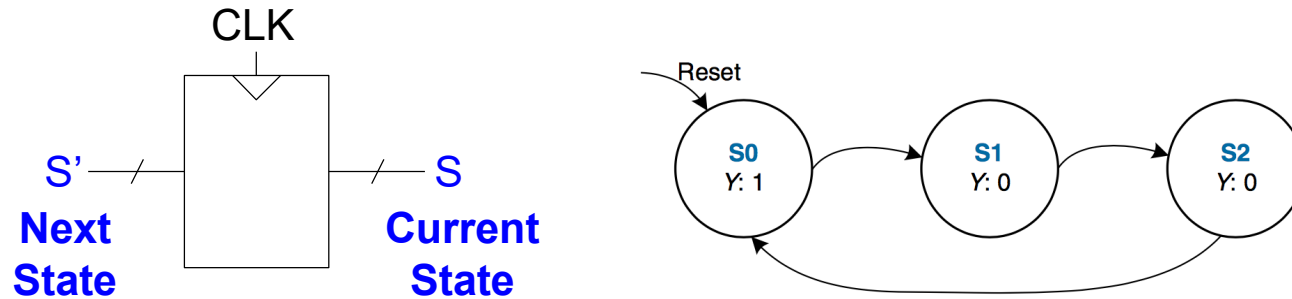


Implementing FSM Example 1: Definitions

```
module divideby3FSM (input clk,  
                    input reset,  
                    output y);  
  
    reg [1:0] state, nextstate;  
  
    parameter S0 = 2'b00;  
    parameter S1 = 2'b01;  
    parameter S2 = 2'b10;
```

- We define **state** and **nextstate** as 2-bit reg
- The parameter descriptions are **optional**, it makes reading easier

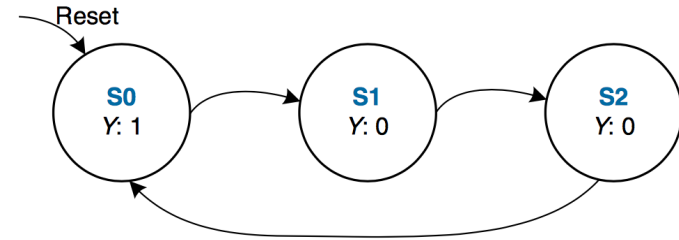
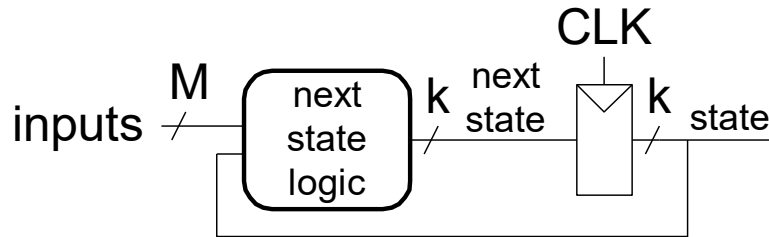
Implementing FSM Example 1: State Register



```
// state register
always @ (posedge clk, posedge reset)
  if (reset) state <= S0;
  else      state <= nextstate;
```

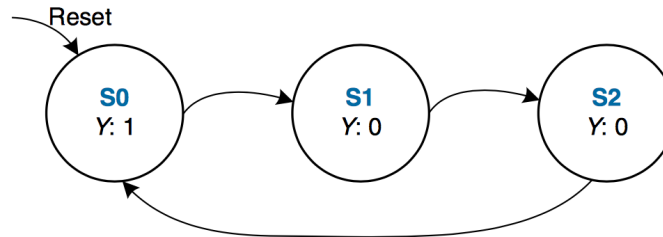
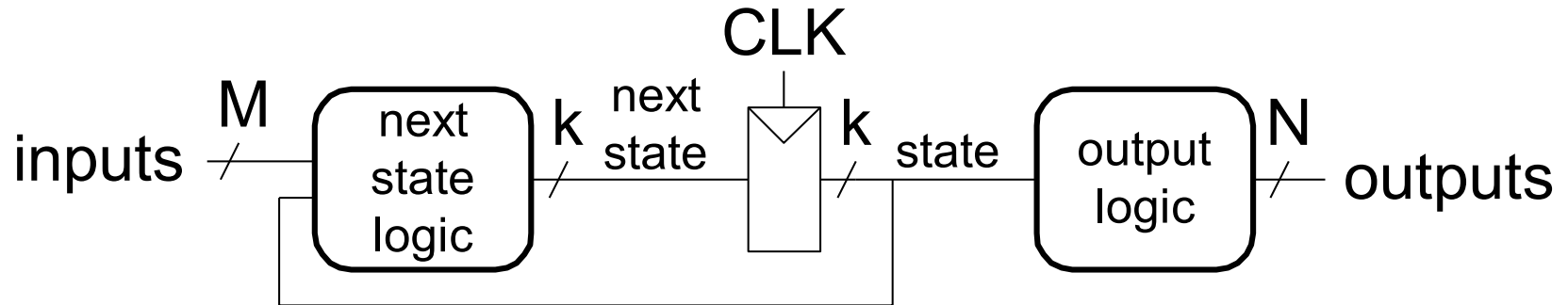
- This part defines the **state register** (memorizing process)
- Sensitive to only **clk**, **reset**
- In this example, **reset** is active when it is '1' (active-high)

Implementing FSM Example 1: Next State Logic



```
// next state logic
always @ (*)
  case (state)
    S0:      nextstate = S1;
    S1:      nextstate = S2;
    S2:      nextstate = S0;
    default: nextstate = S0;
  endcase
```

Implementing FSM Example 1: Output Logic



```
// output logic
assign y = (state == S0);
```

- In this example, output depends only on state
 - **Moore type FSM**

Full Implementation of FSM Example 1

```
module divideby3FSM (input clk, input reset, output y);
    reg [1:0] state, nextstate;

    parameter S0 = 2'b00; parameter S1 = 2'b01; parameter S2 = 2'b10;

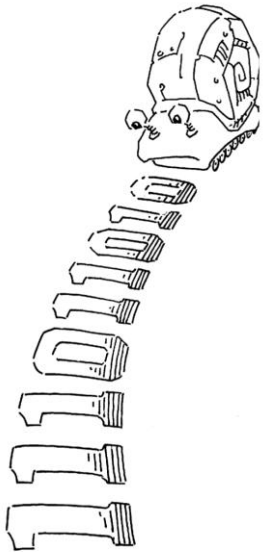
    always @ (posedge clk, posedge reset) // state register
        if (reset) state <= S0;
        else      state <= nextstate;

    always @ (*) // next state logic
        case (state)
            S0:      nextstate = S1;
            S1:      nextstate = S2;
            S2:      nextstate = S0;
            default: nextstate = S0;
        endcase

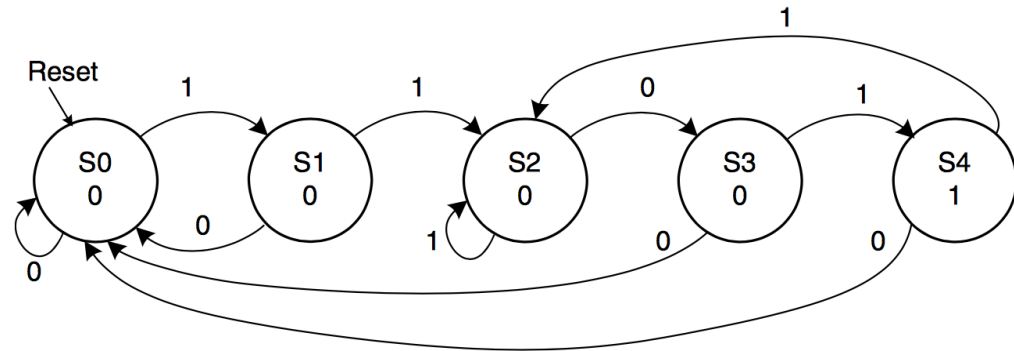
    assign y = (state == S0); // output logic
endmodule
```

FSM Example 2: Recall the Smiling Snail

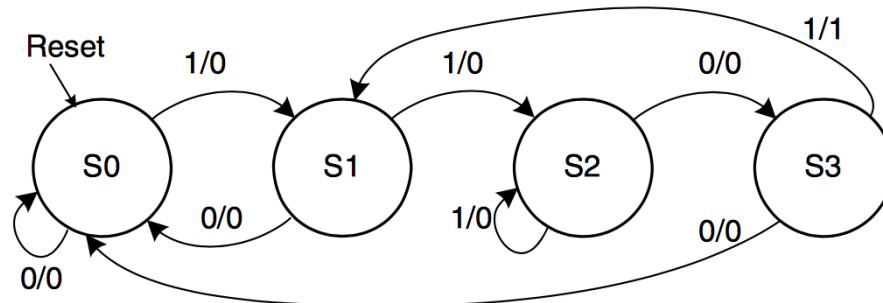
- Alyssa P. Hacker has a snail that crawls down a paper tape with 1's and 0's on it
- The snail smiles whenever the last four digits it has crawled over are **1101**
- Design Moore and Mealy FSMs of the snail's brain



Moore



Mealy



Implementing FSM Example 2: Definitions

```
module SmilingSnail (input clk,  
                    input reset,  
                    input digit,  
                    output smile);
```

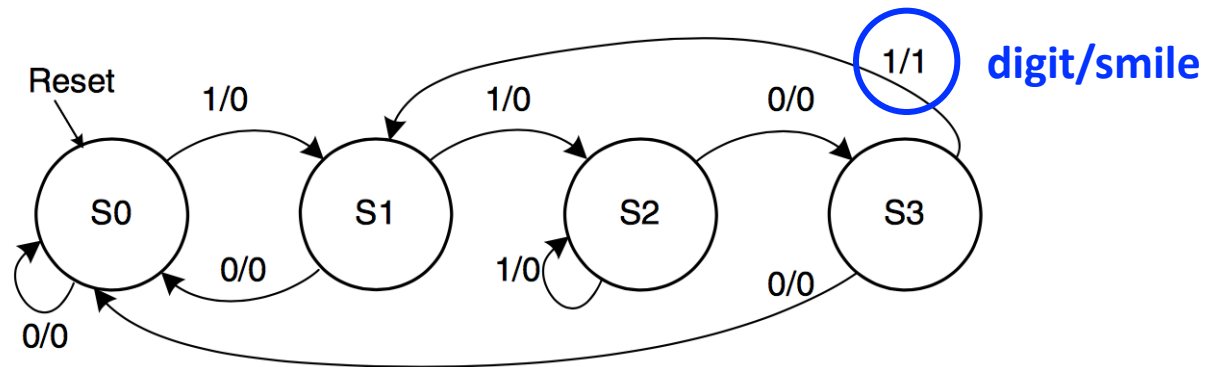
```
    reg [1:0] state, nextstate;
```

```
    parameter S0 = 2'b00;
```

```
    parameter S1 = 2'b01;
```

```
    parameter S2 = 2'b10;
```

```
    parameter S3 = 2'b11;
```



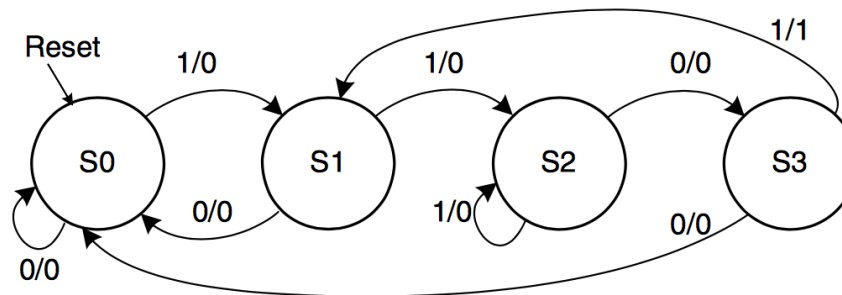
Implementing FSM Example 2: State Register

```
// state register
always @ (posedge clk, posedge reset)
    if (reset) state <= S0;
    else      state <= nextstate;
```

- This part defines the **state register**
- Sensitive to only **clk**, **reset**
- In this example **reset** is active when '1' (active-high)

Implementing FSM Example 2: Next State Logic

```
// next state logic
always @ (*)
  case (state)
    S0: if (digit) nextstate = S1;
        else nextstate = S0;
    S1: if (digit) nextstate = S2;
        else nextstate = S0;
    S2: if (digit) nextstate = S2;
        else nextstate = S3;
    S3: if (digit) nextstate = S1;
        else nextstate = S0;
    default: nextstate = S0;
  endcase
```



Implementing FSM Example 2: Output Logic

```
// output logic  
assign smile = (digit & state == S3);
```

- In this example, output depends on state and input
 - **Mealy type FSM**
- We used a simple combinational assignment

Implementation of FSM Example 2

```
module SmilingSnail (input clk,
                    input reset,
                    input digit,
                    output smile);

    reg [1:0] state, nextstate;

    parameter S0 = 2'b00;
    parameter S1 = 2'b01;
    parameter S2 = 2'b10;
    parameter S3 = 2'b11;

    // state register
    always @ (posedge clk, posedge
reset)
        if (reset) state <= S0;
        else      state <= nextstate;
```

```
    always @ (*) // next state logic
        case (state)
            S0: if (digit)
                    nextstate = S1;
                else nextstate = S0;
            S1: if (digit)
                    nextstate = S2;
                else nextstate = S0;
            S2: if (digit)
                    nextstate = S2;
                else nextstate = S3;
            S3: if (digit)
                    nextstate = S1;
                else nextstate = S0;
            default: nextstate = S0;
        endcase
    // output logic
    assign smile = (digit & state==S3);

endmodule
```

What Did We Learn?

- Basics of describing **sequential circuits** in Verilog
- The **always** statement
 - ❑ Needed for describing memorizing elements (**flip-flops, latches**)
 - ❑ Can also be used to describe **combinational circuits**
- **Blocking** vs **Non-blocking** statements
 - ❑ **=** assigns the value **immediately**
 - ❑ **<=** assigns the value **at the end of the block**
- **Describing FSMs in Verilog**
 - ❑ Next state logic
 - ❑ State assignment
 - ❑ Output logic

Timing and Verification

What Will We Learn Today?

- **Timing** in **combinational circuits**
 - ❑ Propagation delay and contamination delay
 - ❑ Glitches
- **Timing** in **sequential circuits**
 - ❑ Setup time and hold time
 - ❑ Determining how fast a circuit can operate
- **Circuit Verification**
 - ❑ How to make sure a circuit works correctly
 - ❑ Functional verification
 - ❑ Timing verification

Tradeoffs in Circuit Design

Circuit Design is a Tradeoff Between:

■ Area

- Circuit **area** is proportional to the **cost** of the device

■ Speed (Latency, Throughput)

- We want **faster**, more **capable** circuits

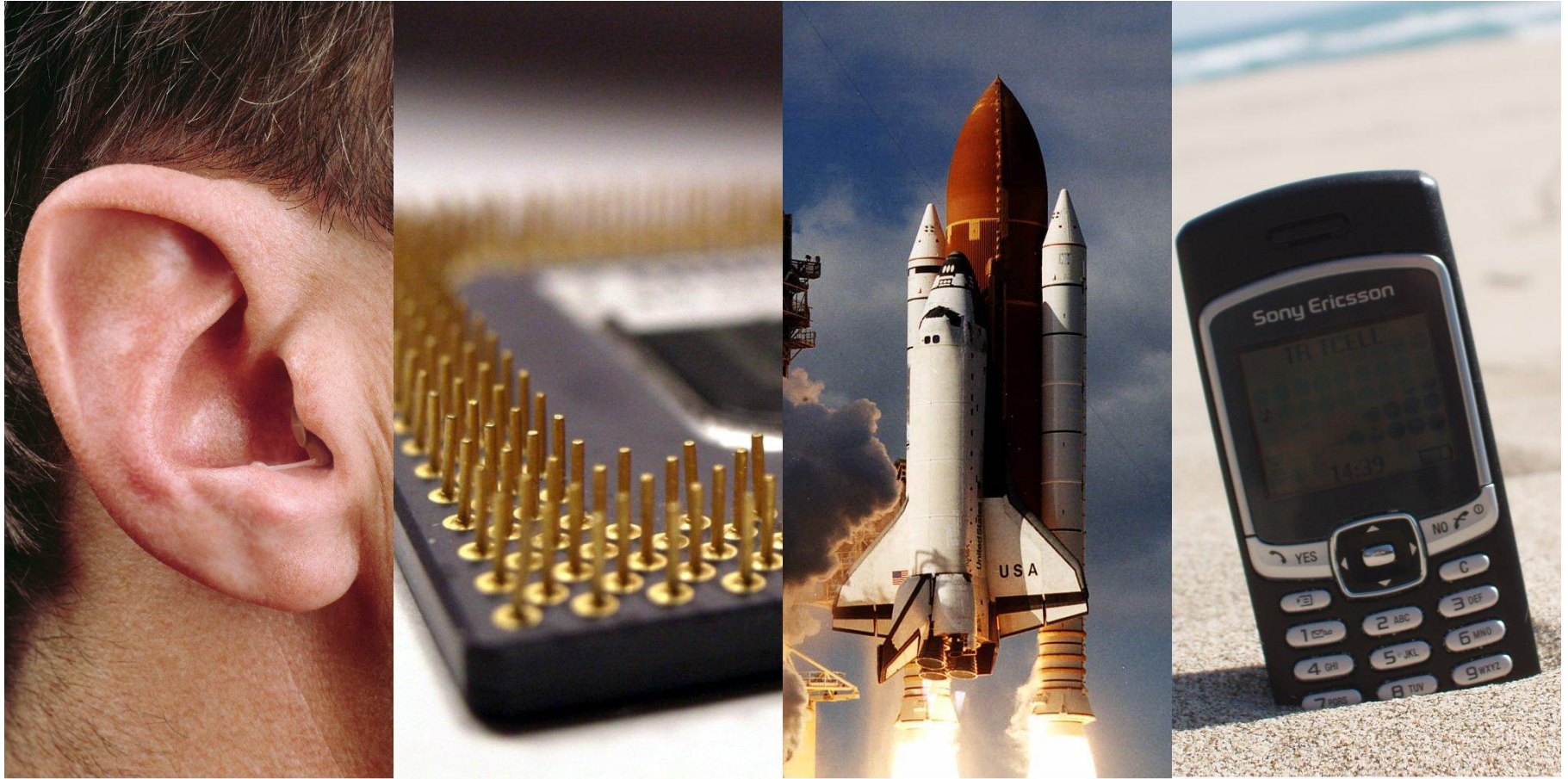
■ Power & Energy Consumption

- Mobile devices need to work with a **limited** power supply
- High performance devices **dissipate** more than 100W/cm²

■ Design Time

- Designers are **expensive** in *time* and *money*
- The **competition** will not wait for you

Requirements and Goals Depend On Application



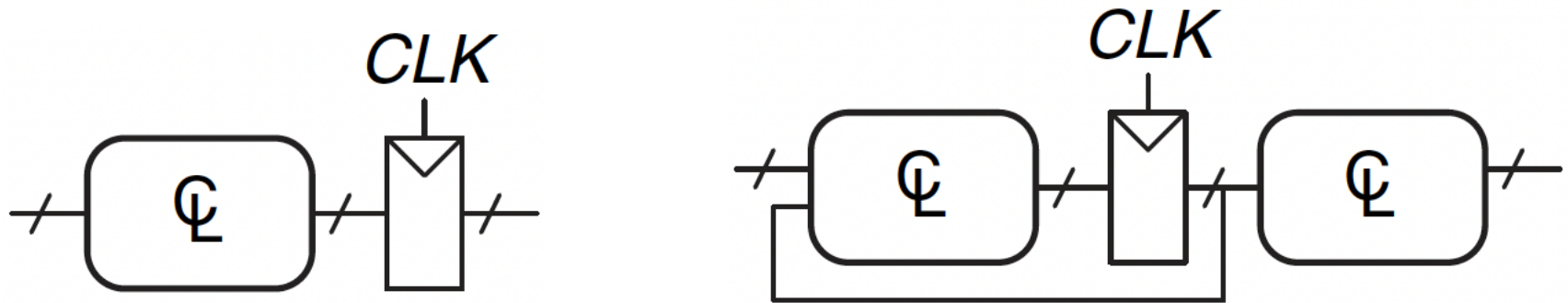
Circuit Timing

- Until now, we investigated **logical functionality**
- What about **timing**?
 - How **fast** is a circuit?
 - How can we make a circuit **faster**?
 - What happens if we run a circuit **too fast**?
- A design that is logically correct can still **fail** because of real-world **implementation issues**!

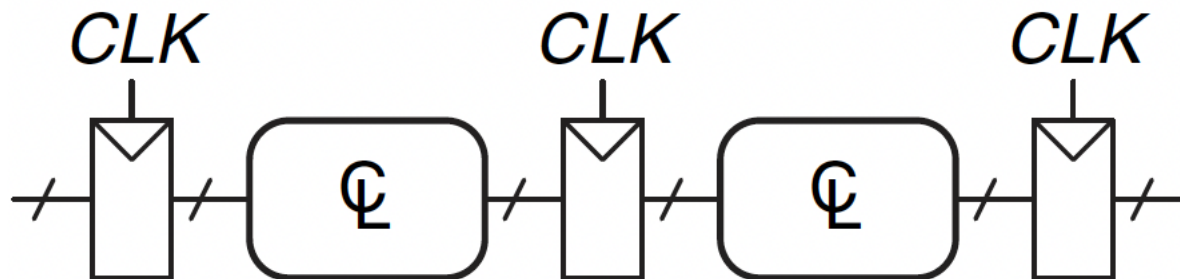
Part 1:

Combinational Circuit Timing

Combinational & Sequential Circuits

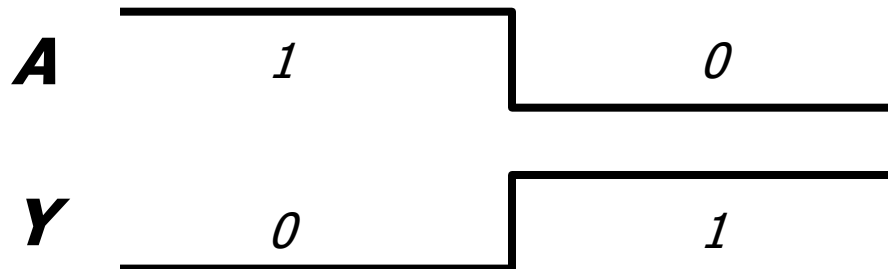
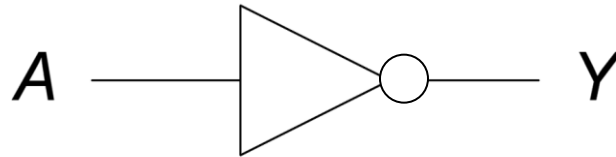


Major types of Sequential Logic Circuits



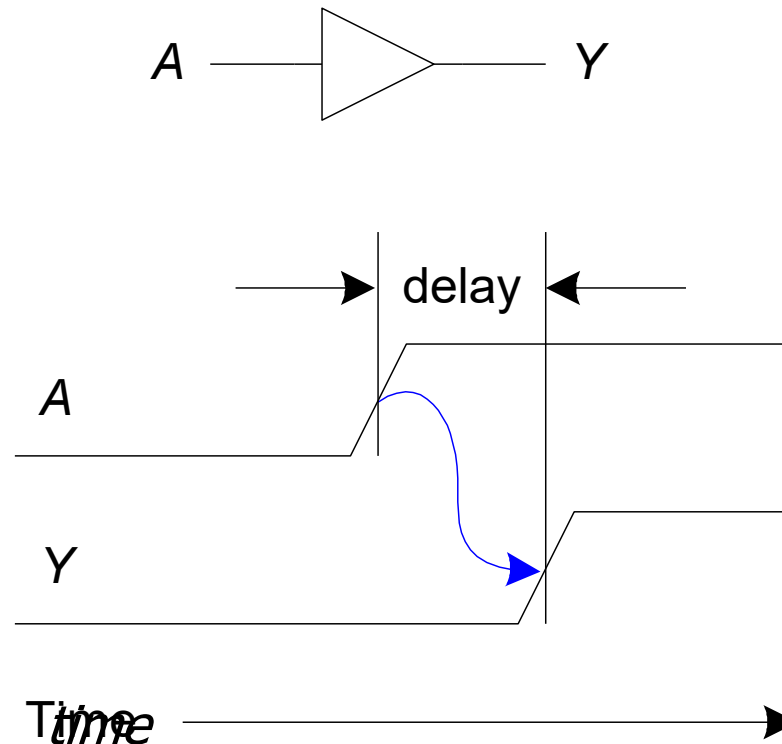
Digital Logic Abstraction

- “**Digital logic**” is a convenient **abstraction**
 - Output changes *immediately* with the input

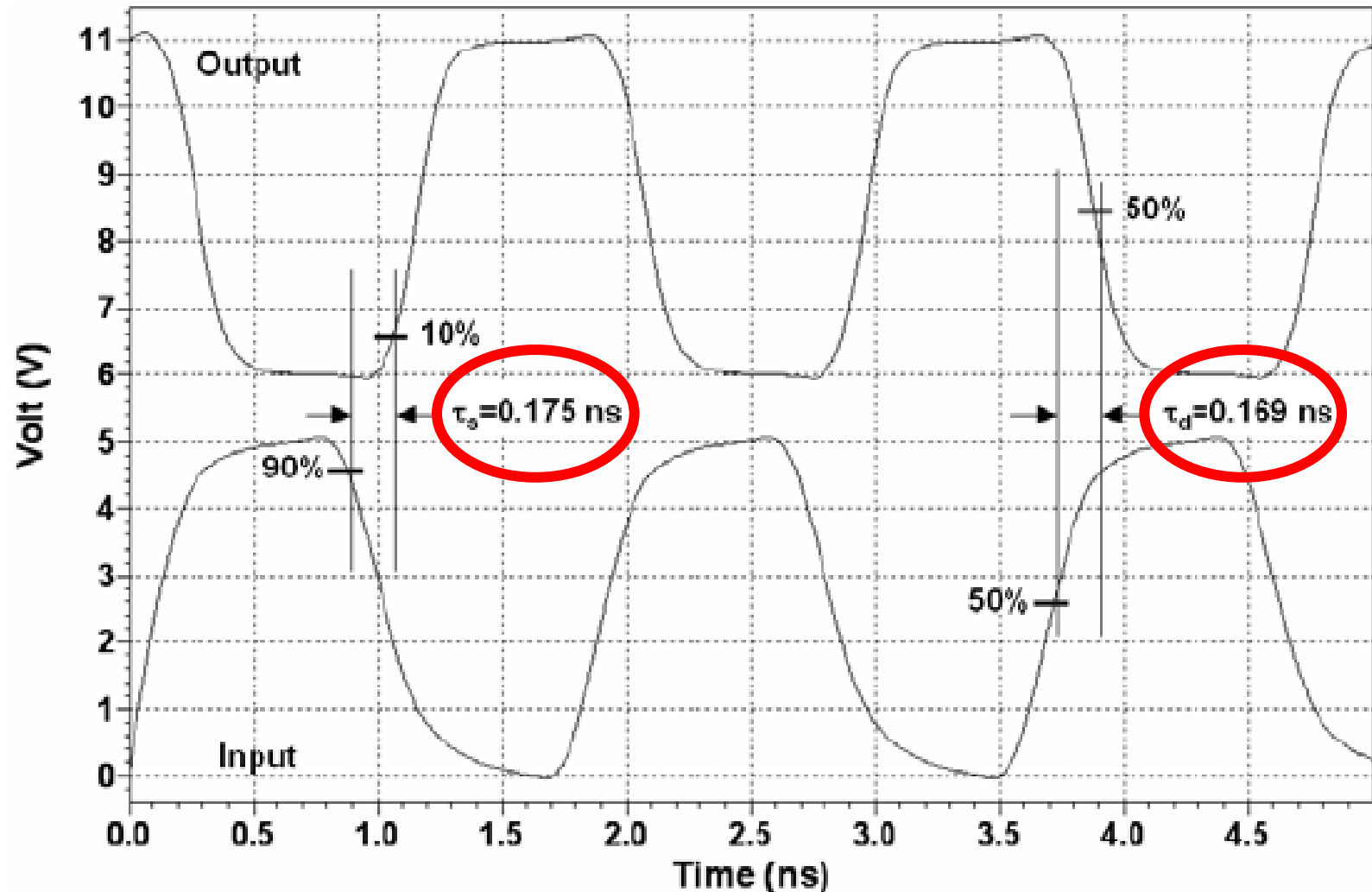


Combinational Circuit Delay

- In reality, **outputs** are **delayed** from **inputs**
 - Transistors take a finite amount of time to switch



Real Inverter Delay Example



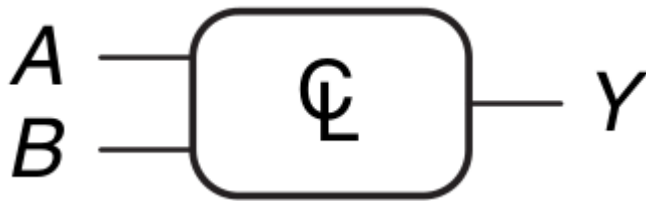
Circuit Delay and Its Variation

- Delay is fundamentally caused by
 - **Capacitance** and **resistance** in a circuit
 - Finite **speed of light** (not so fast on a nanosecond scale!)
- **Anything** affecting these quantities can change delay:
 - **Rising** (i.e., $0 \rightarrow 1$) vs. **falling** (i.e., $1 \rightarrow 0$) inputs
 - Different **input values** lead to different **delays**
 - Changes in **environment** (e.g., temperature)
 - **Aging** of the circuit
 - **Voltage** supplied to the circuit
- We have a **range of possible delays** from input to output

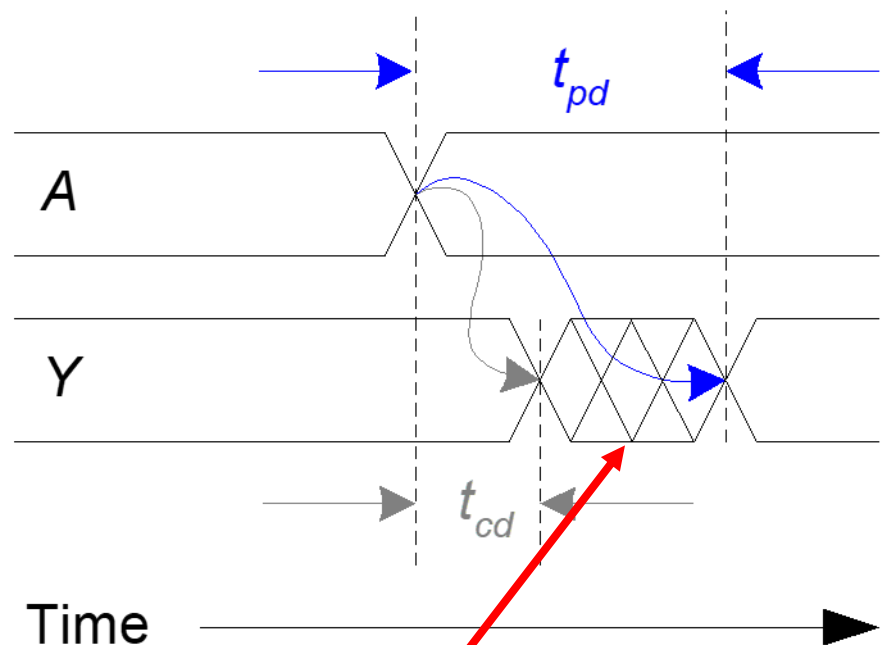
Delays from Input to Output Y

- **Contamination delay (t_{cd}):** delay until Y ***starts changing***
- **Propagation delay (t_{pd}):** delay until Y ***finishes changing***

Example Circuit



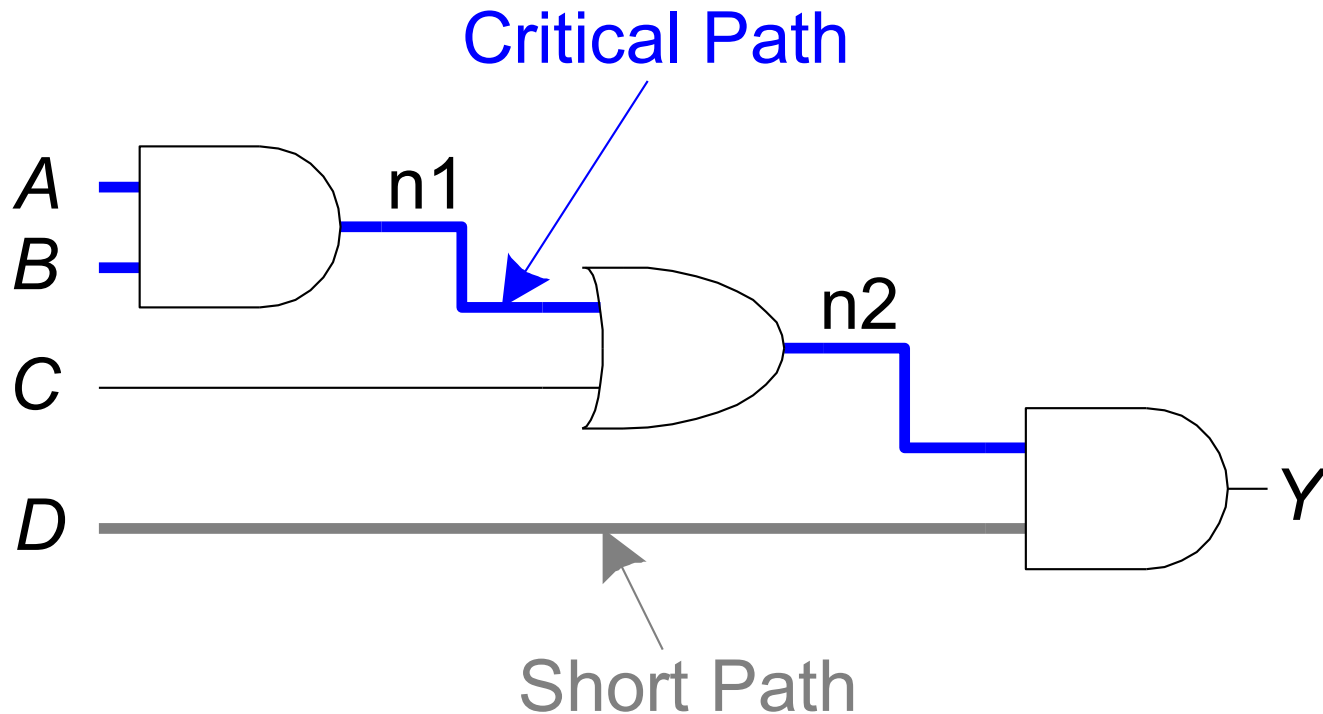
Effect of Changing Input 'A'



**Cross-hatching
means value is changing**

Calculating Longest & Shortest Delay Paths

- We care about **both** the *longest* and *shortest* delay paths in a circuit (we will see why later in the lecture)



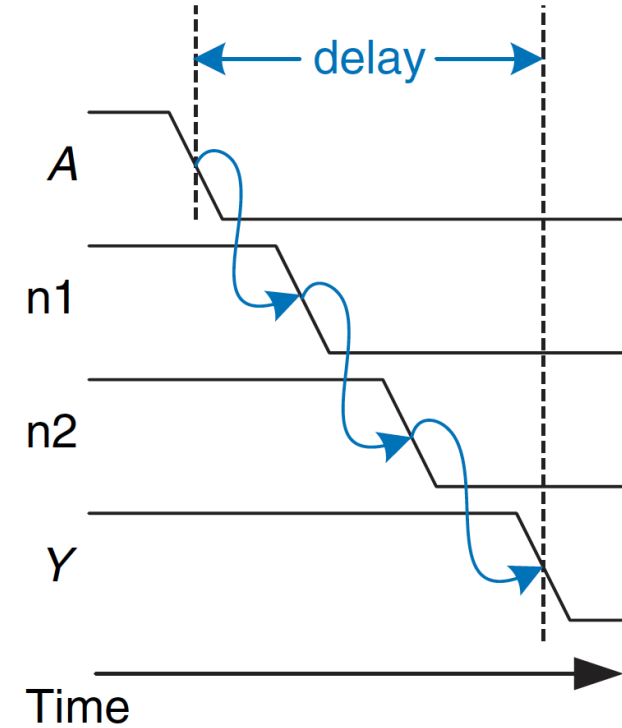
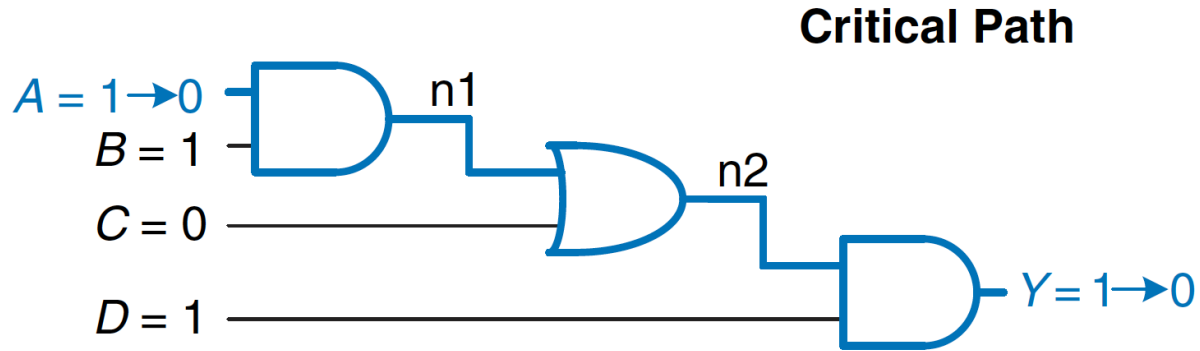
- **Critical (Longest) Path:**

$$t_{pd} = 2 t_{pd_AND} + t_{pd_OR}$$

- **Shortest Path:**

$$t_{cd} = t_{cd_AND}$$

Calculating Longest Delay Path (Critical Path)



- **Critical (Longest) Path:**

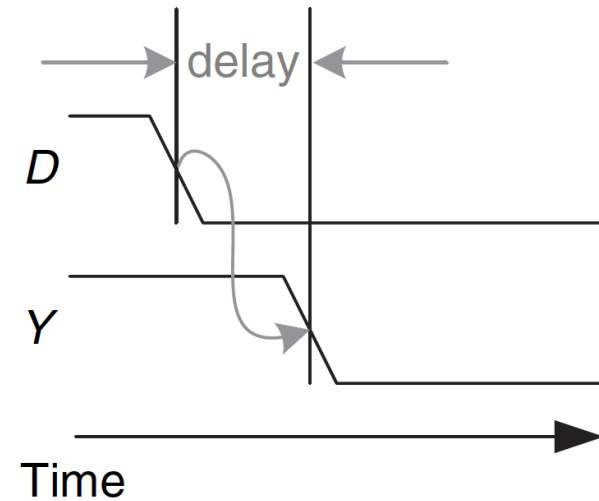
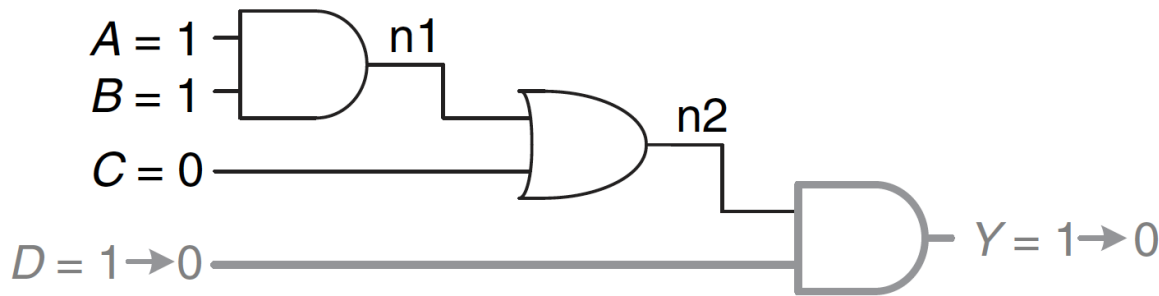
$$t_{pd} = 2 t_{pd_AND} + t_{pd_OR}$$

- **Shortest Path:**

$$t_{cd} = t_{cd_AND}$$

Calculating Shortest Delay Path

Short Path



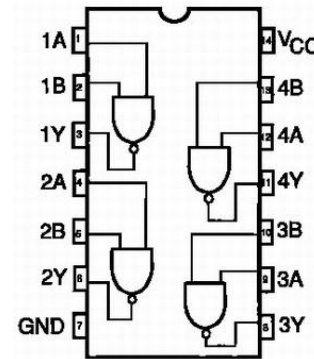
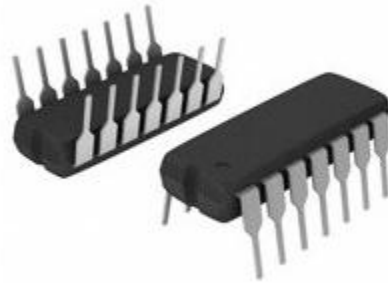
- **Critical (Longest) Path:**

$$t_{pd} = 2 t_{pd_AND} + t_{pd_OR}$$

- **Shortest Path:**

$$t_{cd} = t_{cd_AND}$$

Example t_{pd} for a Real NAND-2 Gate



Symbol	Parameter	Conditions	25 °C			−40 °C to +125 °C		Unit
			Min	Typ	Max	Max (85 °C)	Max (125 °C)	
74HC00								
t _{pd}	propagation delay	nA, nB to nY; see Figure 6 [1]						
		V _{CC} = 2.0 V	-	25	-	115	135	ns
		V _{CC} = 4.5 V	-	9	-	23	27	ns
		V _{CC} = 5.0 V; C _L = 15 pF	-	7	-	-	-	ns
		V _{CC} = 6.0 V	-	7	-	20	23	ns

- Heavy **dependence** on **voltage** and **temperature**!

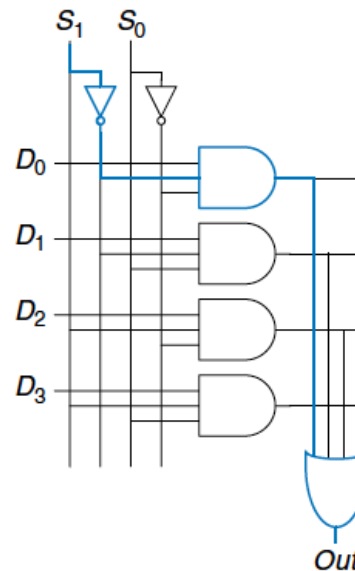
Example Worst-Case t_{pd}

- Two different **implementations** of a **4:1 multiplexer**

Gate Delays

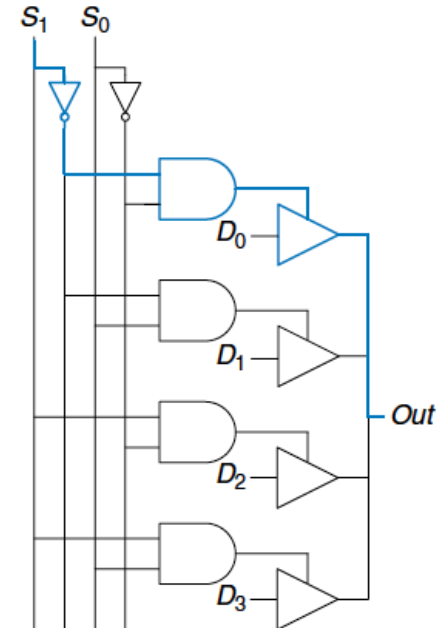
Gate	t_{pd} (ps)
NOT	30
2-input AND	60
3-input AND	80
4-input OR	90
tristate (A to Y)	50
tristate (enable to Y)	35

Implementation 1



$$\begin{aligned} t_{pd_sy} &= t_{pd_INV} + t_{pd_AND3} + t_{pd_OR4} \\ &= 30 \text{ ps} + 80 \text{ ps} + 90 \text{ ps} \\ &= 200 \text{ ps} \\ \text{(a)} \quad t_{pd_dy} &= t_{pd_AND3} + t_{pd_OR4} \\ &= 170 \text{ ps} \end{aligned}$$

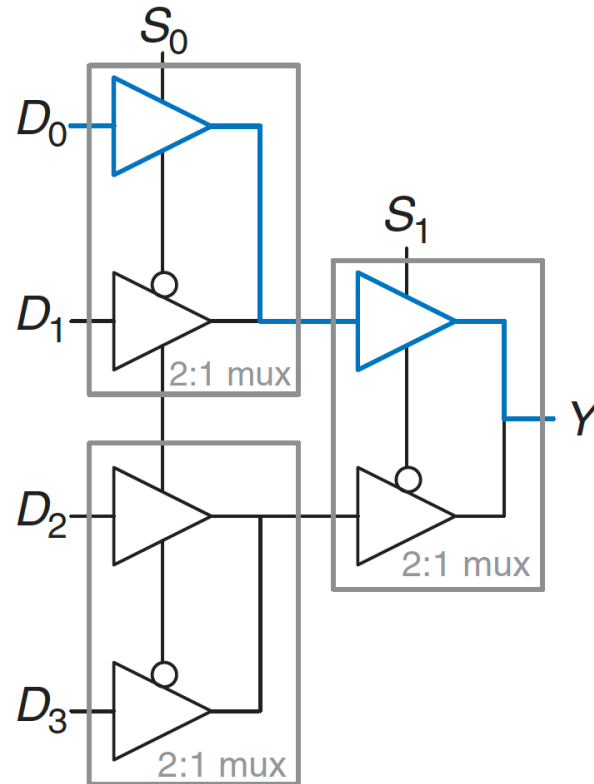
Implementation 2



$$\begin{aligned} t_{pd_sy} &= t_{pd_INV} + t_{pd_AND2} + t_{pd_TRI_SY} \\ &= 30 \text{ ps} + 60 \text{ ps} + 35 \text{ ps} \\ &= 125 \text{ ps} \\ \text{(b)} \quad t_{pd_dy} &= t_{pd_TRI_AY} \\ &= 50 \text{ ps} \end{aligned}$$

- Different designs lead to very **different delays**

Aside: A Third 4:1 MUX Implementation



$$t_{pd_{s0y}} = t_{pd_TRLSY} + t_{pd_TRI_AY} = 85 \text{ ns}$$

$$t_{pd_{dy}} = 2 t_{pd_TRI_AY} = 100 \text{ ns}$$

Figure 2.74 4:1 multiplexer propagation delays: hierarchical using 2:1 multiplexers

Disclaimer: Calculating Long/Short Paths

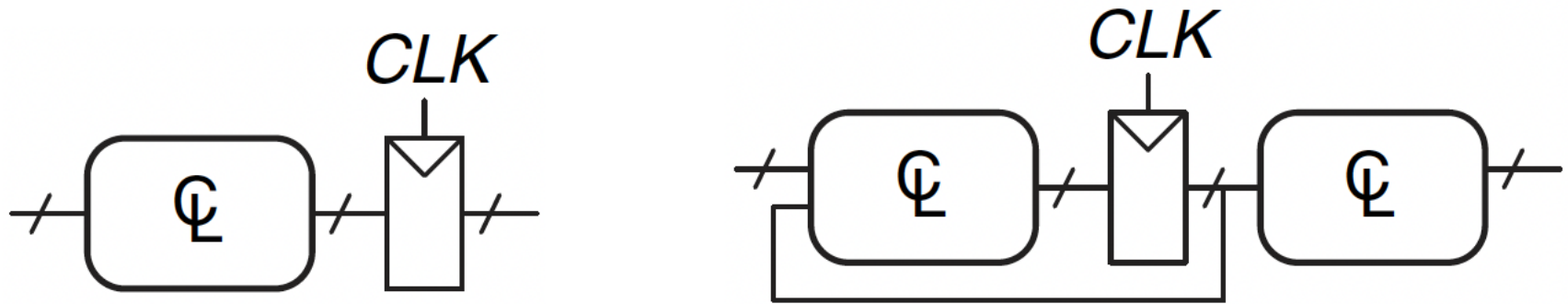
- It's **not** always this easy to determine the long/short paths!
 - Not all **input transitions** affect the **output**
 - Can have **multiple different paths** from an input to output
- In reality, circuits are **not** all built equally
 - Different instances of the **same gate** have **different delays**
 - **Wires** have **nonzero delay** (increasing with length)
 - Temperature/voltage affect circuit speeds
 - Not all circuit elements are affected the same way
 - Can even **change the critical path**!
- Designers assume "**worst-case**" **conditions** and run many **statistical simulations** to balance yield/performance

Combinational Timing Summary

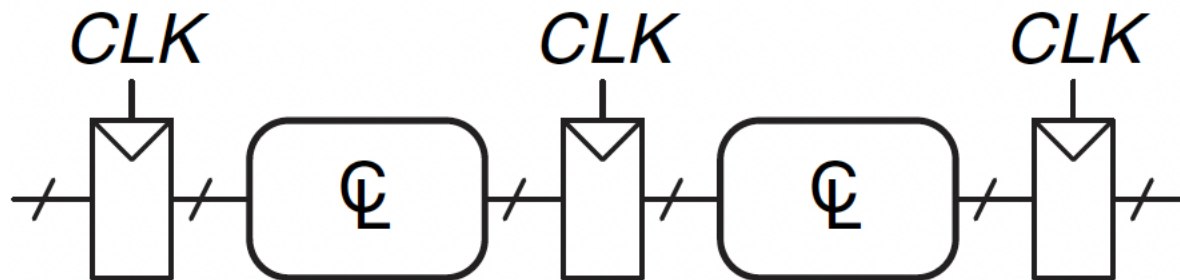
- Circuit outputs change some time **after** the inputs change
 - Caused by finite speed of light (not so fast on a ns scale!)
 - Delay is dependent on inputs, environmental state, etc.
- The range of possible delays is characterized by:
 - **Contamination delay (t_{cd})**: *minimum possible* delay
 - **Propagation delay (t_{pd})**: *maximum possible* delay
- Delays **change** with:
 - Circuit design (e.g., topology, materials)
 - Operating conditions

Output Glitches

Combinational & Sequential Circuits



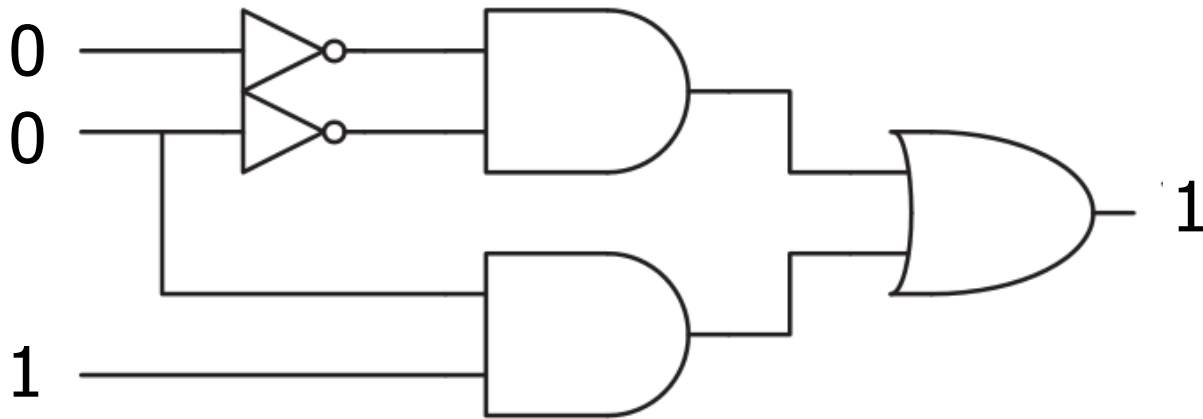
Major types of Sequential Logic Circuits



Glitches

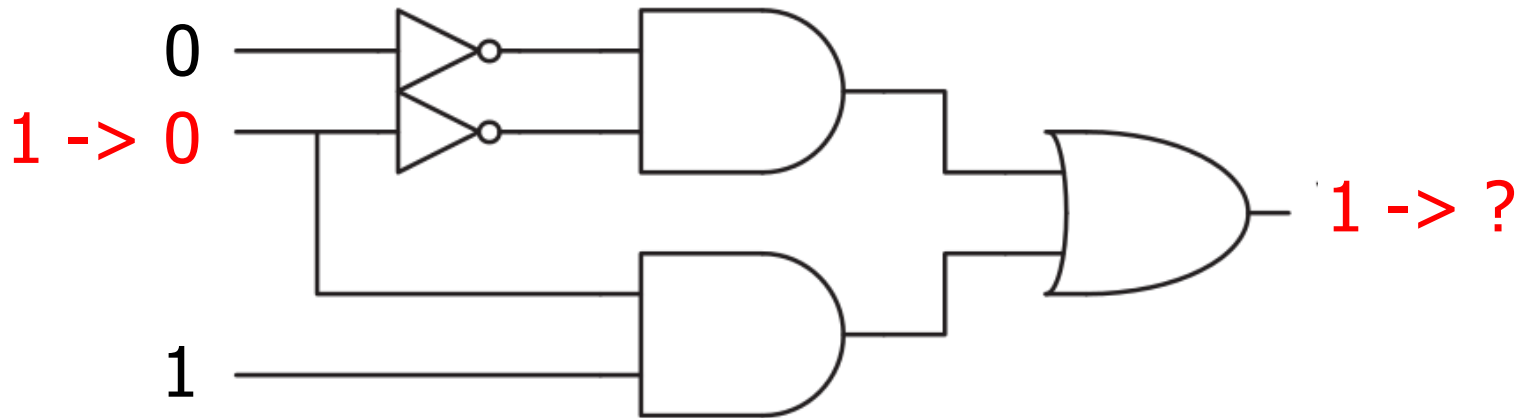
- **Glitch:** **one** input transition causes **multiple** output transitions

Circuit initial state



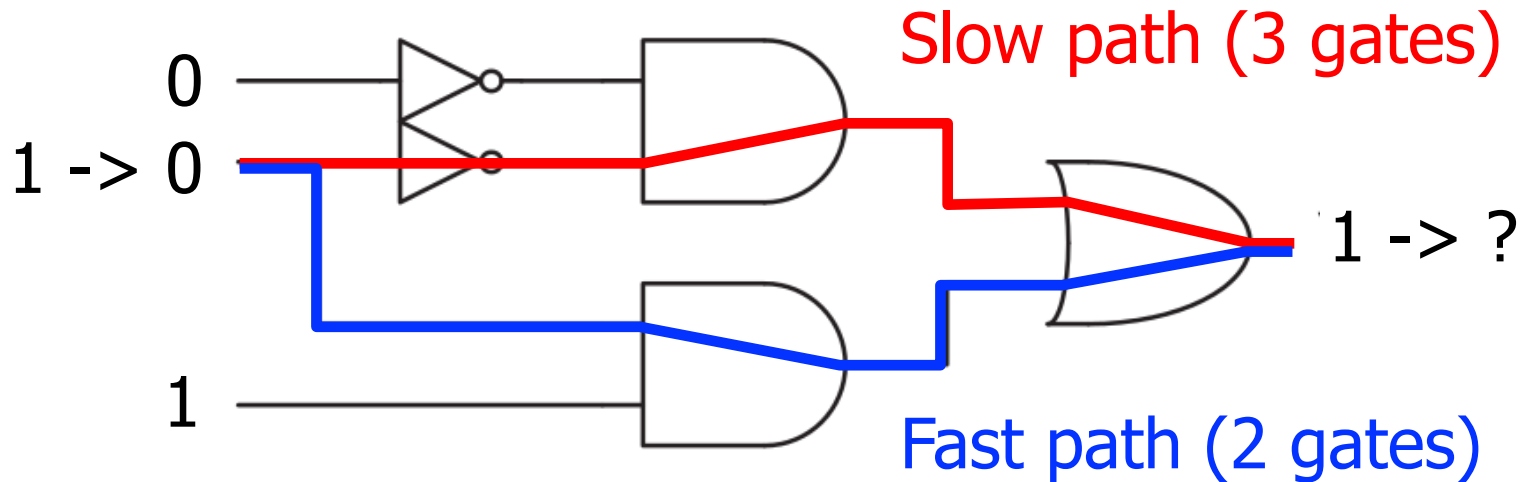
Glitches

- **Glitch:** **one** input transition causes **multiple** output transitions



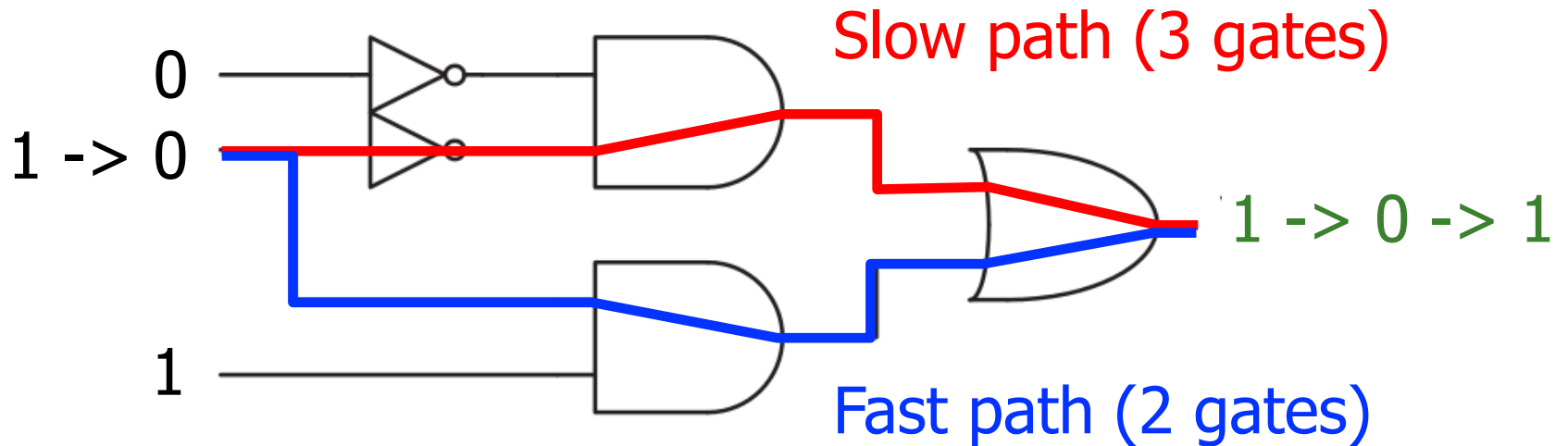
Glitches

- **Glitch:** **one** input transition causes **multiple** output transitions



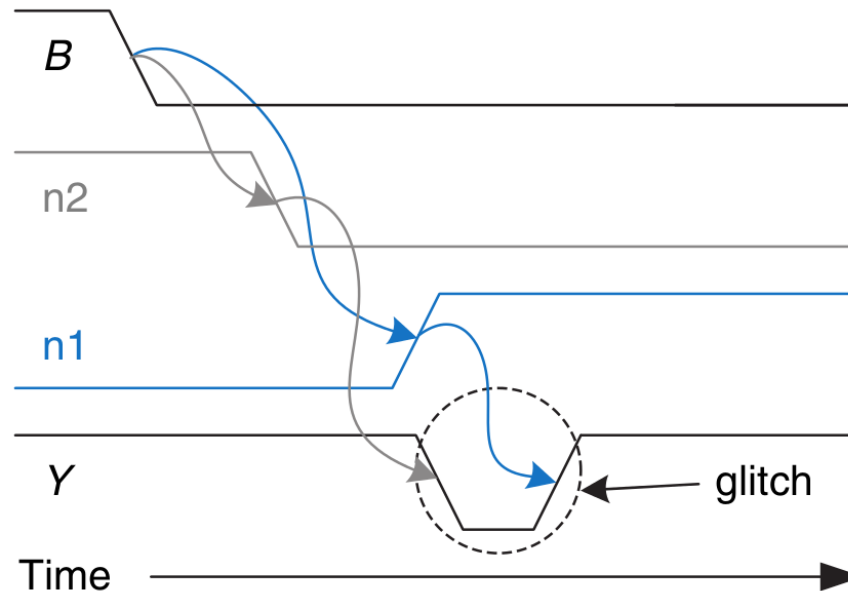
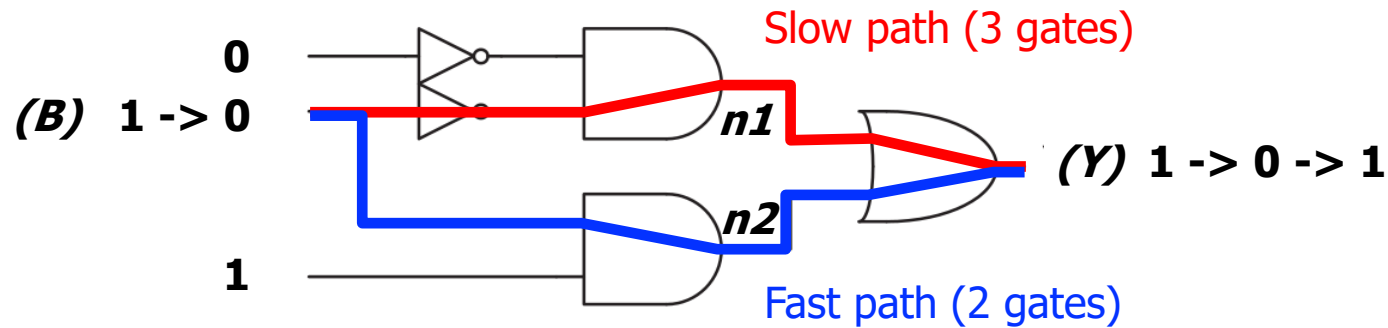
Glitches

- **Glitch:** **one** input transition causes **multiple** output transitions

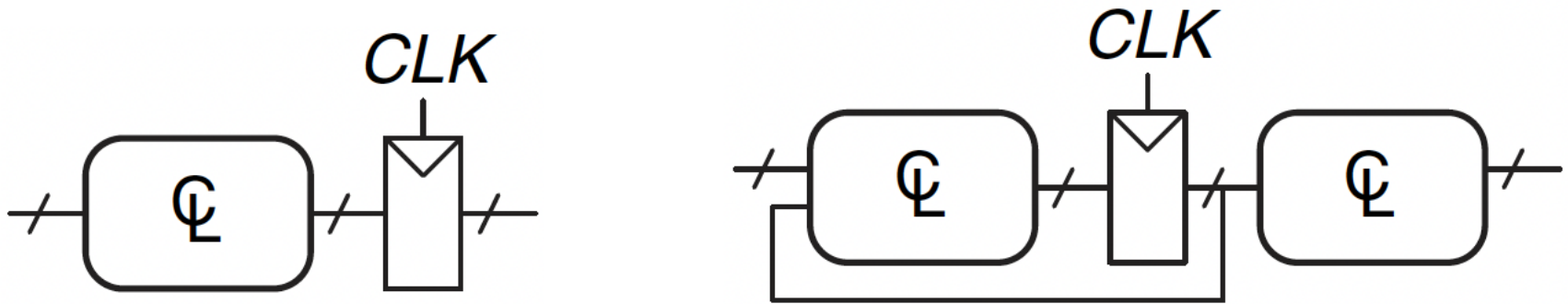


Glitches

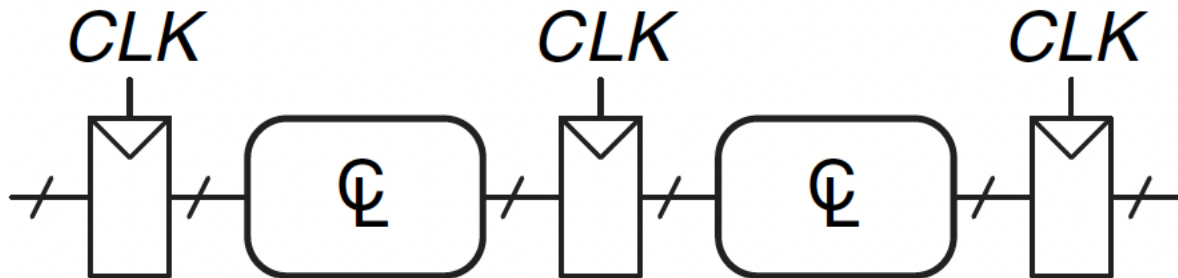
- **Glitch:** one input transition causes **multiple** output transitions



When Could A Glitch Be Bad?

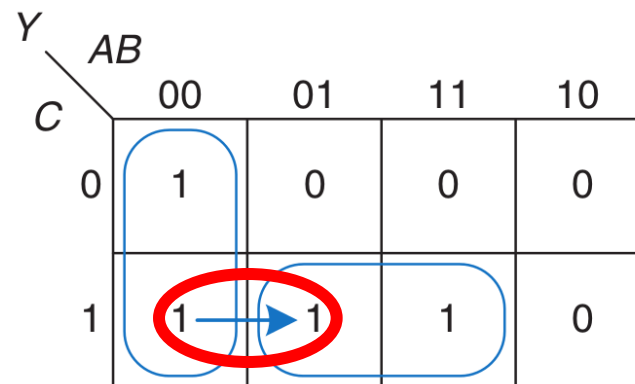
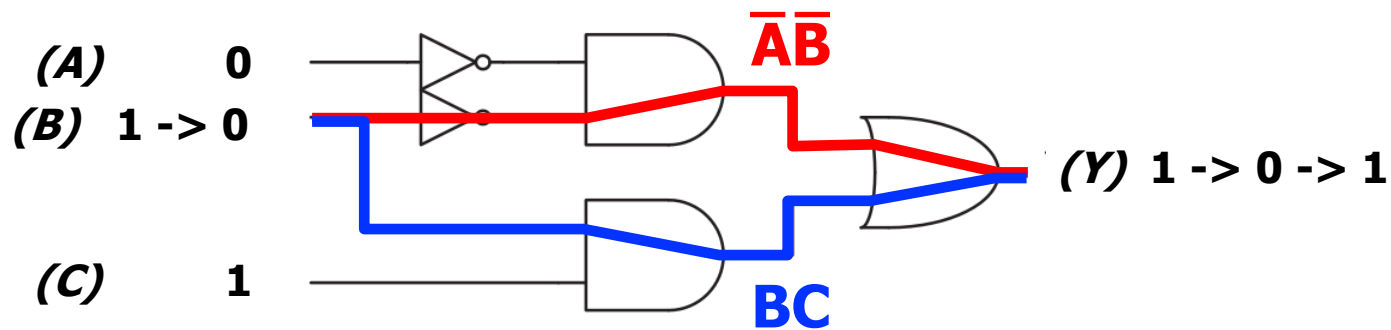


Major types of Sequential Logic Circuits



Optional: Avoiding Glitches Using K-Maps

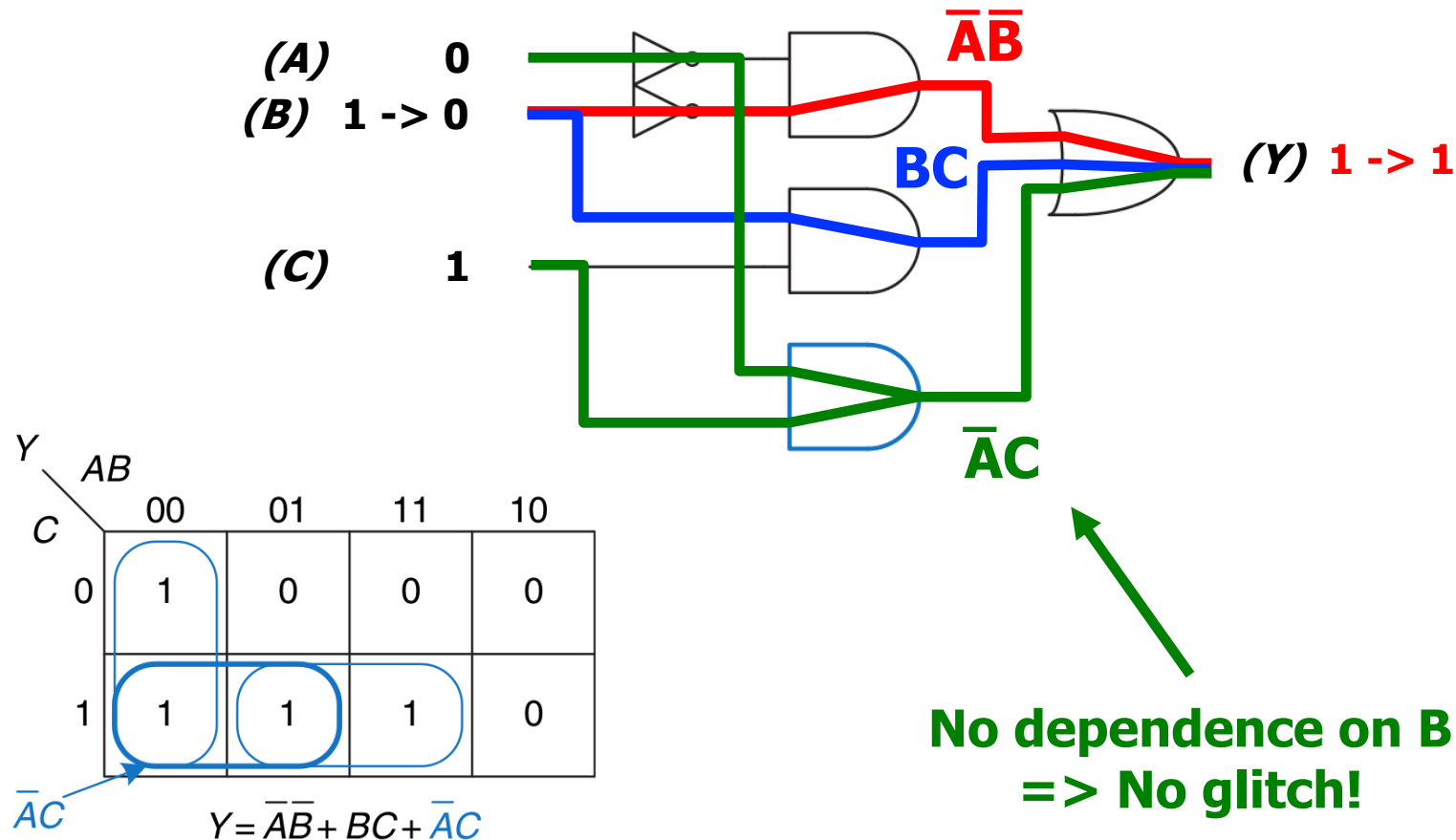
- Glitches are **visible** in **K-maps**
 - Recall: K-maps show the results of a change in a **single input**
 - A glitch occurs when **moving between prime implicants**



$$Y = \overline{AB} + BC$$

Optional: Avoiding Glitches Using K-Maps

- We can **fix** the issue by adding in the **consensus** term
 - Ensures **no transition** between different **prime implicants**



Avoiding Glitches

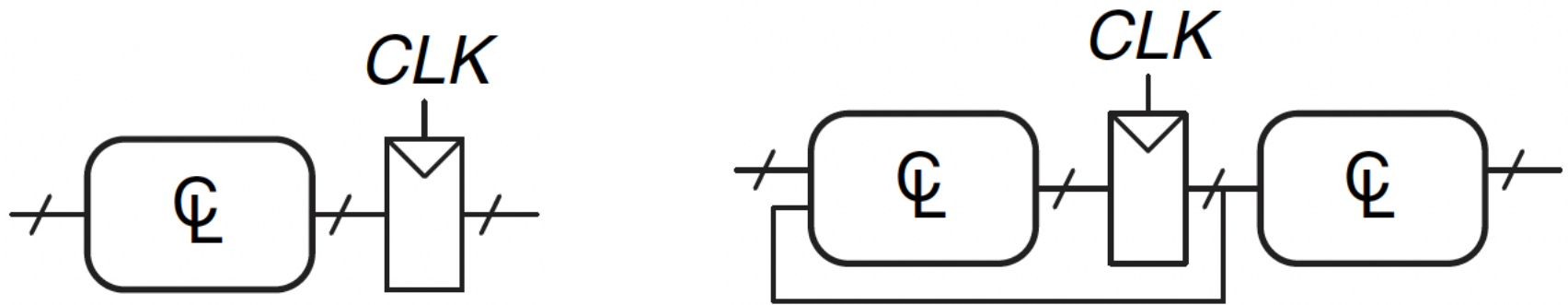
- **Q:** Do we **always** care about glitches?
 - **Fixing** glitches is **undesirable**
 - More chip **area**
 - More **power consumption**
 - More **design effort**
 - The circuit is **eventually** guaranteed to **converge** to the **right value** regardless of glitchiness

- **A:** No, not always!
 - If we only care about the **long-term steady state output**, we can **safely ignore** glitches
 - Up to the **designer to decide** if glitches matter in their application
 - When examining simulation output, important to recognize glitches

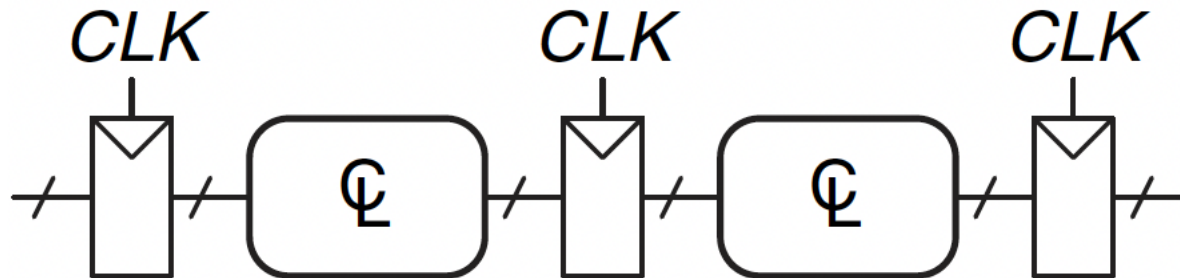
Part 2:

Sequential Circuit Timing

Recall: Synchronous Sequential Circuits

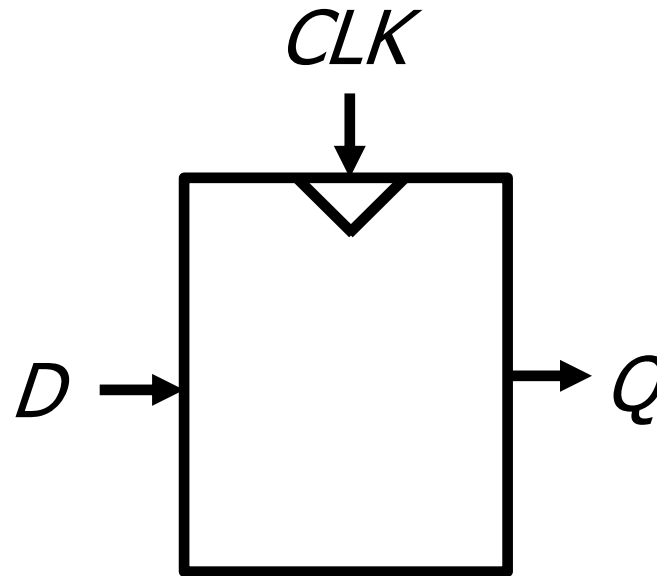


Major types of Sequential Logic Circuits



Recall: D Flip-Flop

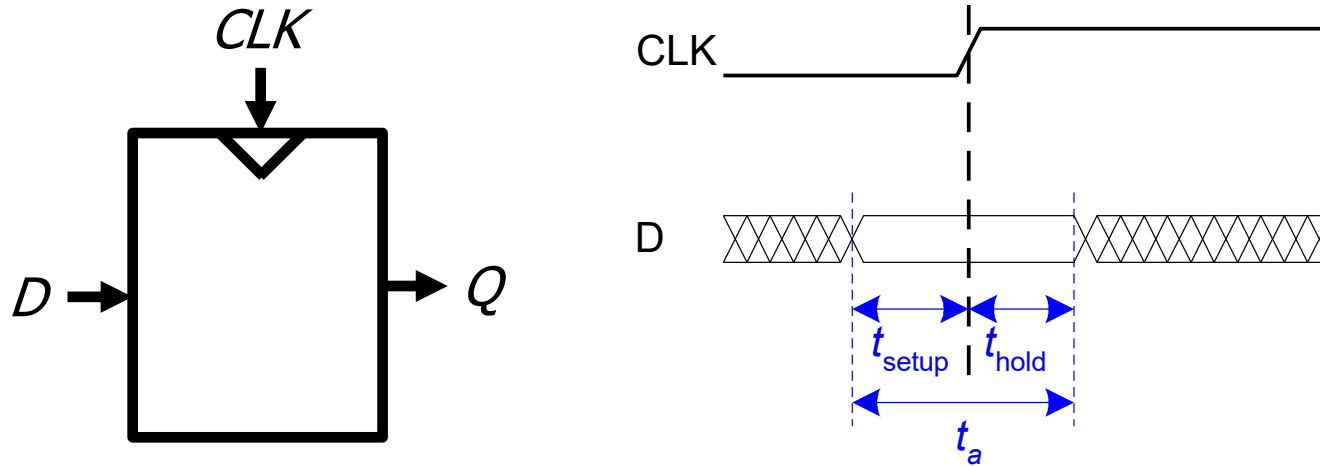
- Flip-flop **samples D** at the **rising clock edge**
 - It outputs the **sampled value** to Q
 - It **"stores"** the **sampled value** until the next rising clock edge



- The D flip-flop is **made** from **combinational** elements
- **D, Q, CLK all have timing requirements!**

D Flip-Flop Input Timing Constraints

- **D** must be **stable** when **sampled** (i.e., at rising clock edge)

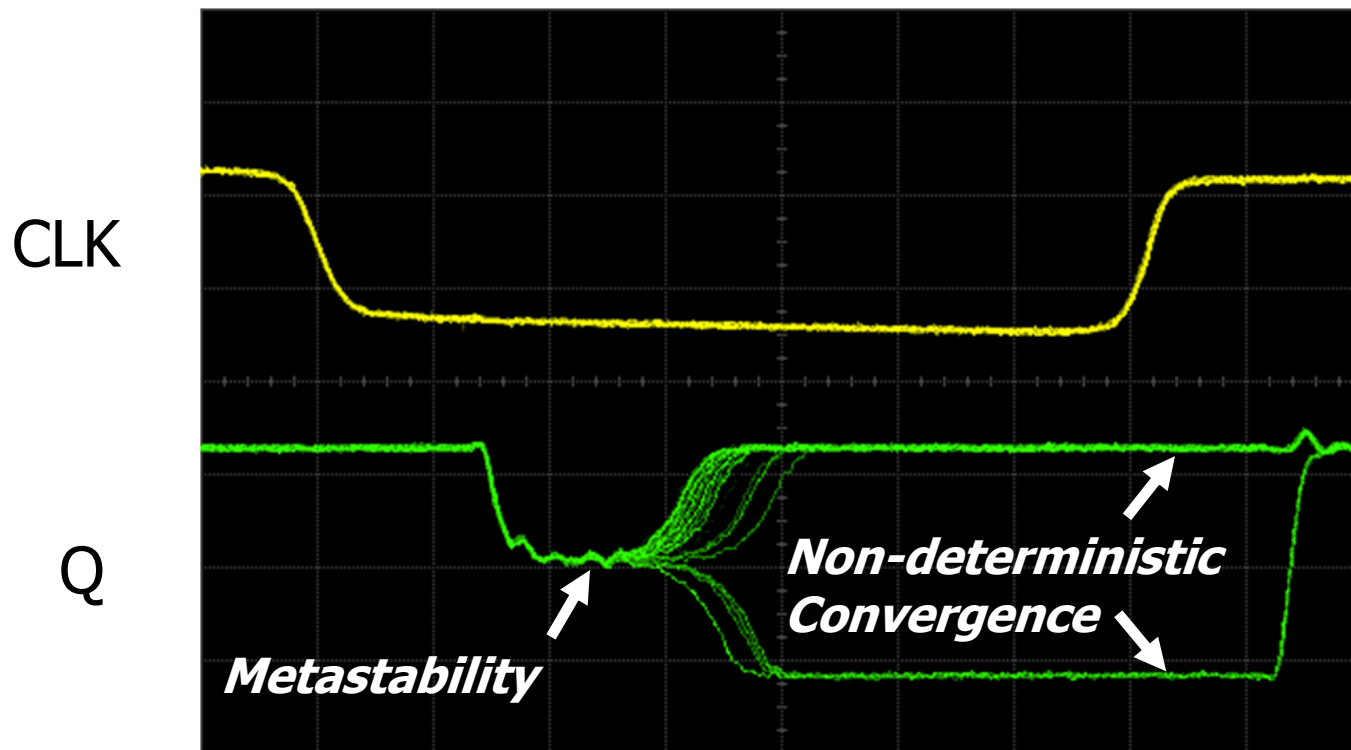


- **Setup time (t_{setup})**: time **before** the clock edge that data must be stable (i.e. not changing)
- **Hold time (t_{hold})**: time **after** the clock edge that data must be stable
- **Aperture time (t_a)**: time **around** clock edge that data must be stable ($t_a = t_{\text{setup}} + t_{\text{hold}}$)

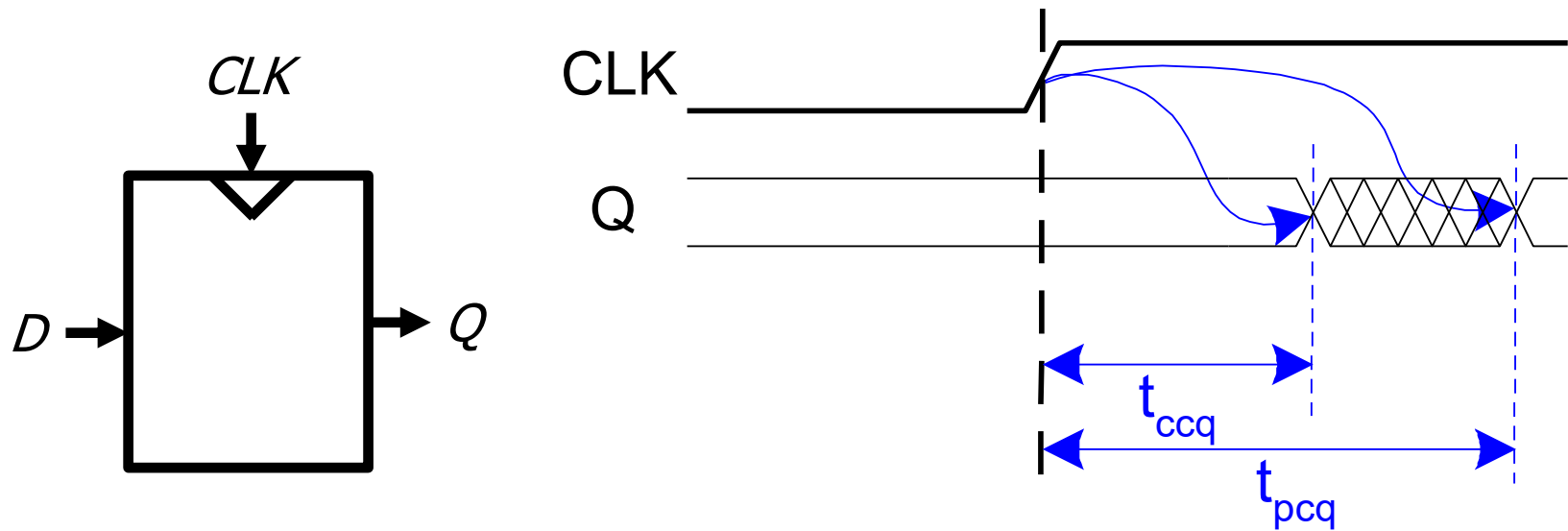
Violating Input Timing: Metastability

- If D is **changing** when sampled, **metastability** can occur
 - ❑ Flip-flop output is **stuck** somewhere between '1' and '0'
 - ❑ Output eventually settles **non-deterministically**

Example Timing Violations (NAND RS Latch)

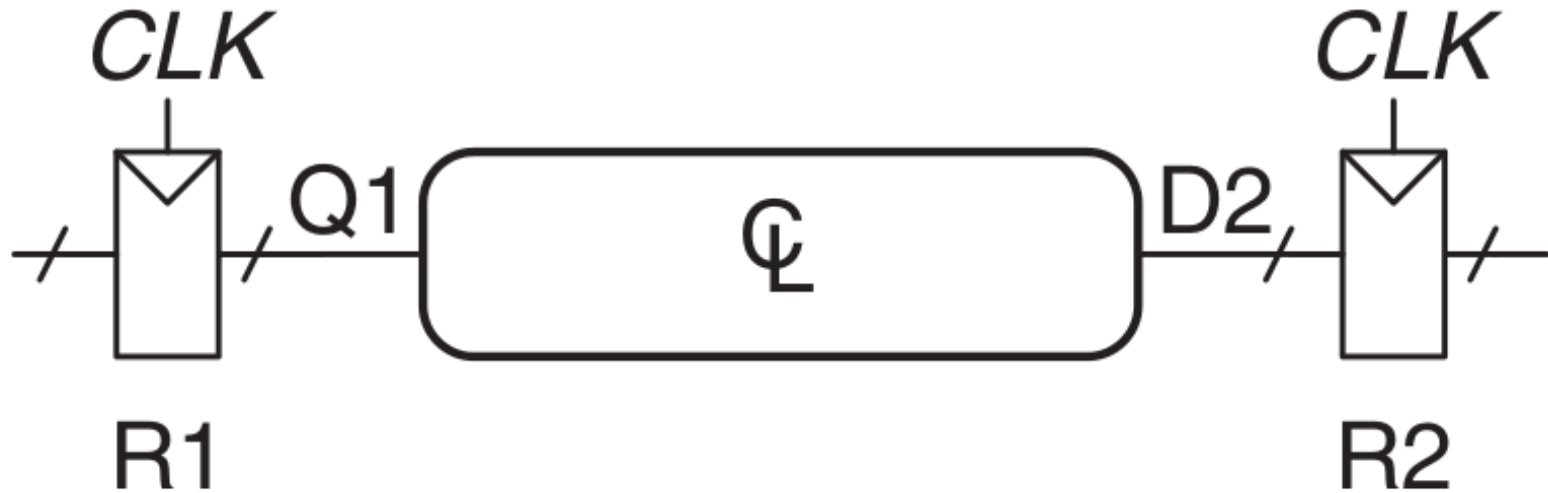


Flip-Flop Output Timing



- **Contamination delay clock-to-q (t_{ccq}):** earliest time after the clock edge that Q starts to change (i.e., is unstable)
- **Propagation delay clock-to-q (t_{pcq}):** latest time after the clock edge that Q stops changing (i.e., is stable)

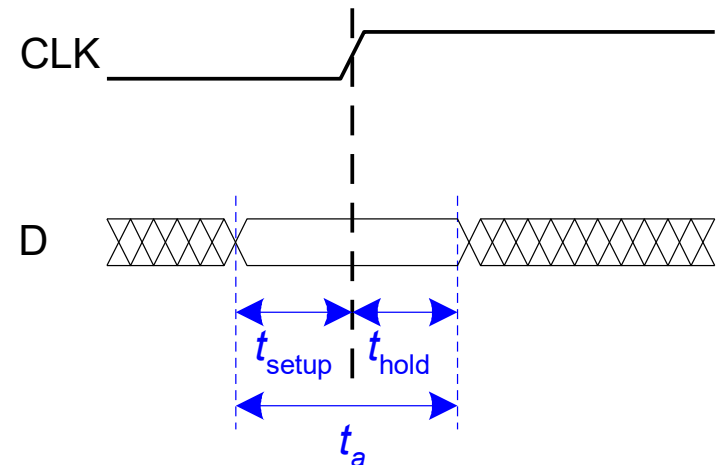
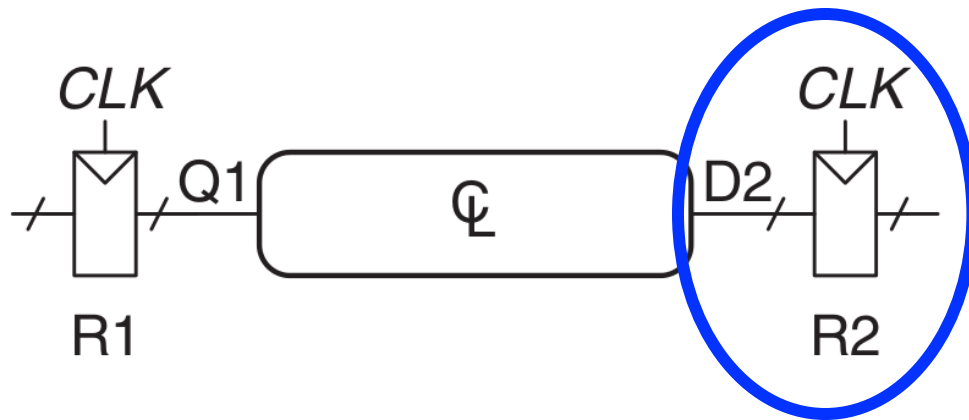
Recall: Sequential System Design



- Multiple **flip-flops** are connected with **combinational logic**
- **Clock** runs with period T_c (cycle time)
- **Must meet timing requirements for both R1 and R2!**

Ensuring Correct Sequential Operation

- Need to ensure correct input timing on **R2**
- Specifically, **D2** must be **stable**:
 - at least t_{setup} **before** the clock edge
 - at least until t_{hold} **after** the clock edge

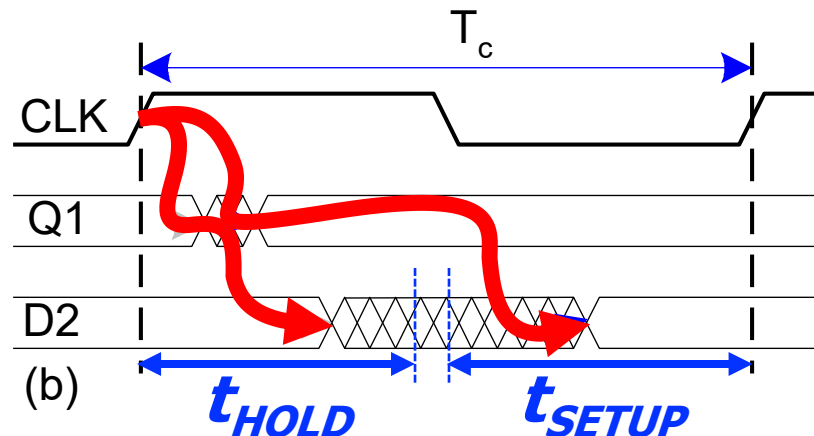
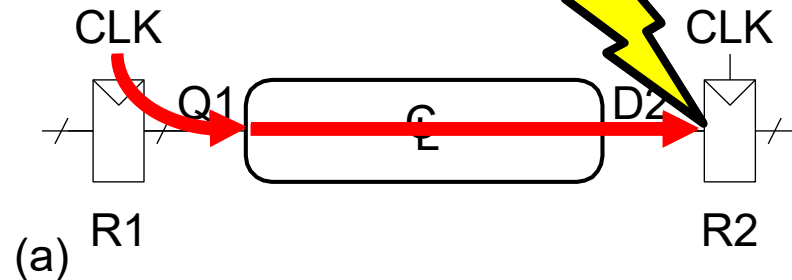


Ensuring Correct Sequential Operation

- This means there is both a **minimum** and **maximum** delay between two flip-flops

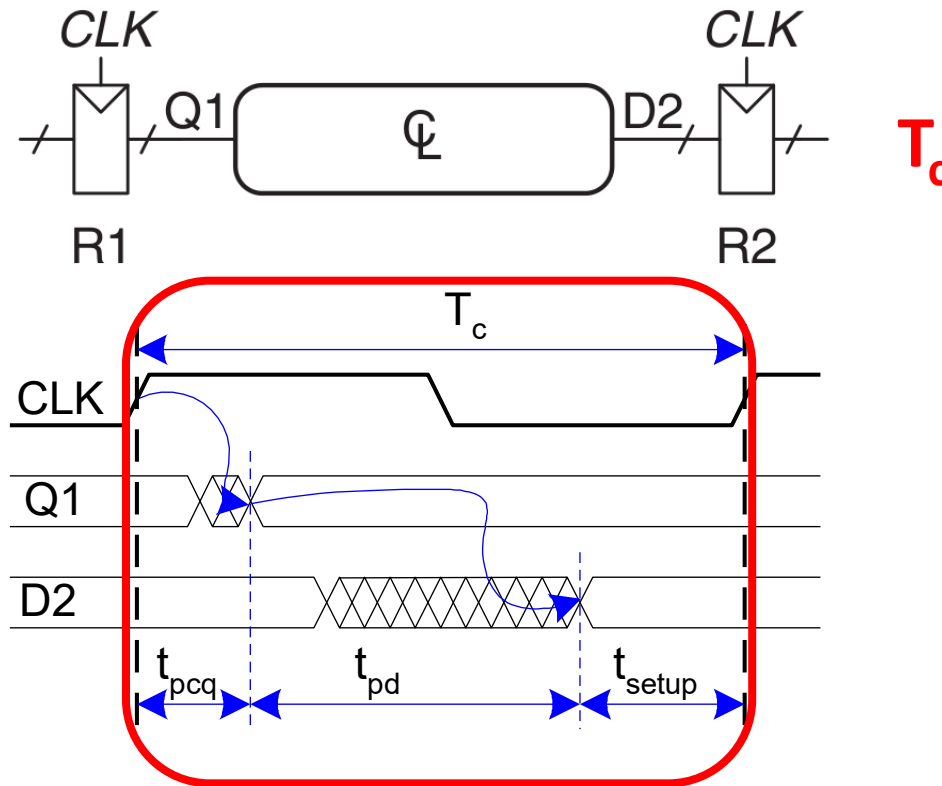
- CL **too fast** -> R2 t_{hold} violation
- CL **too slow** -> R2 t_{setup} violation

**Potential
R2 t_{setup}
VIOLATION!**



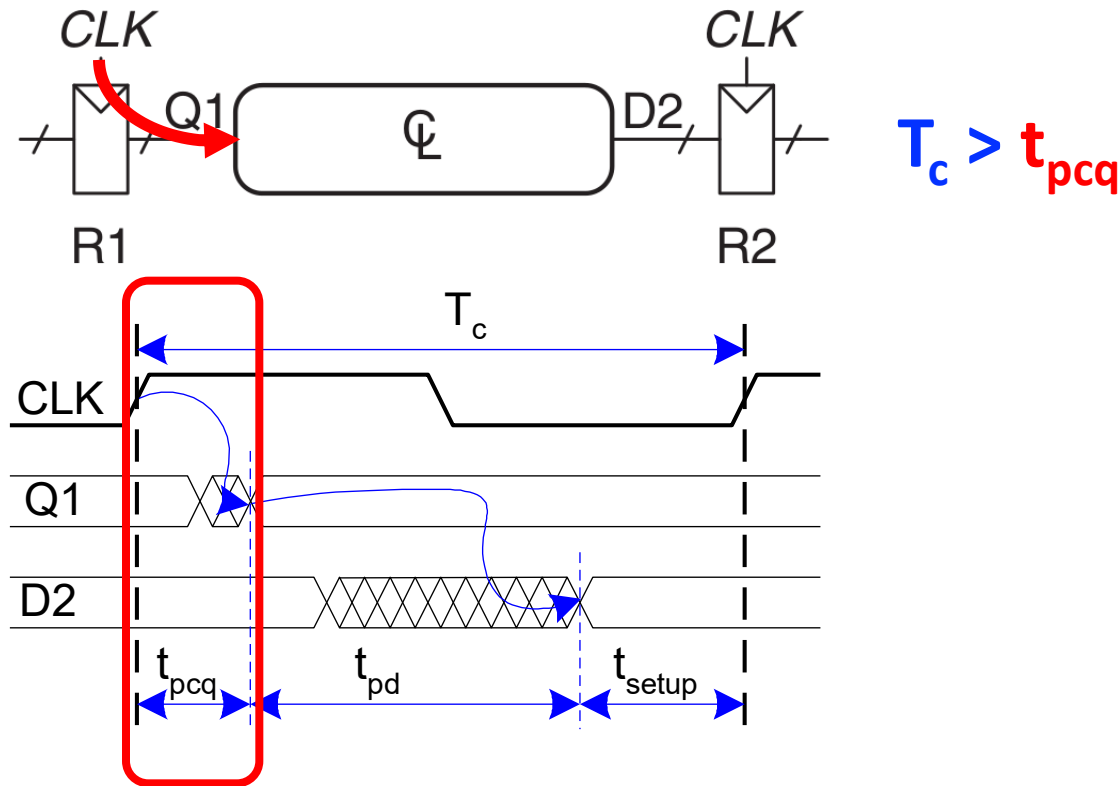
Setup Time Constraint

- **Safe timing** depends on the **maximum** delay from R1 to R2
- The input to R2 must be stable at least t_{setup} **before** the clock edge.



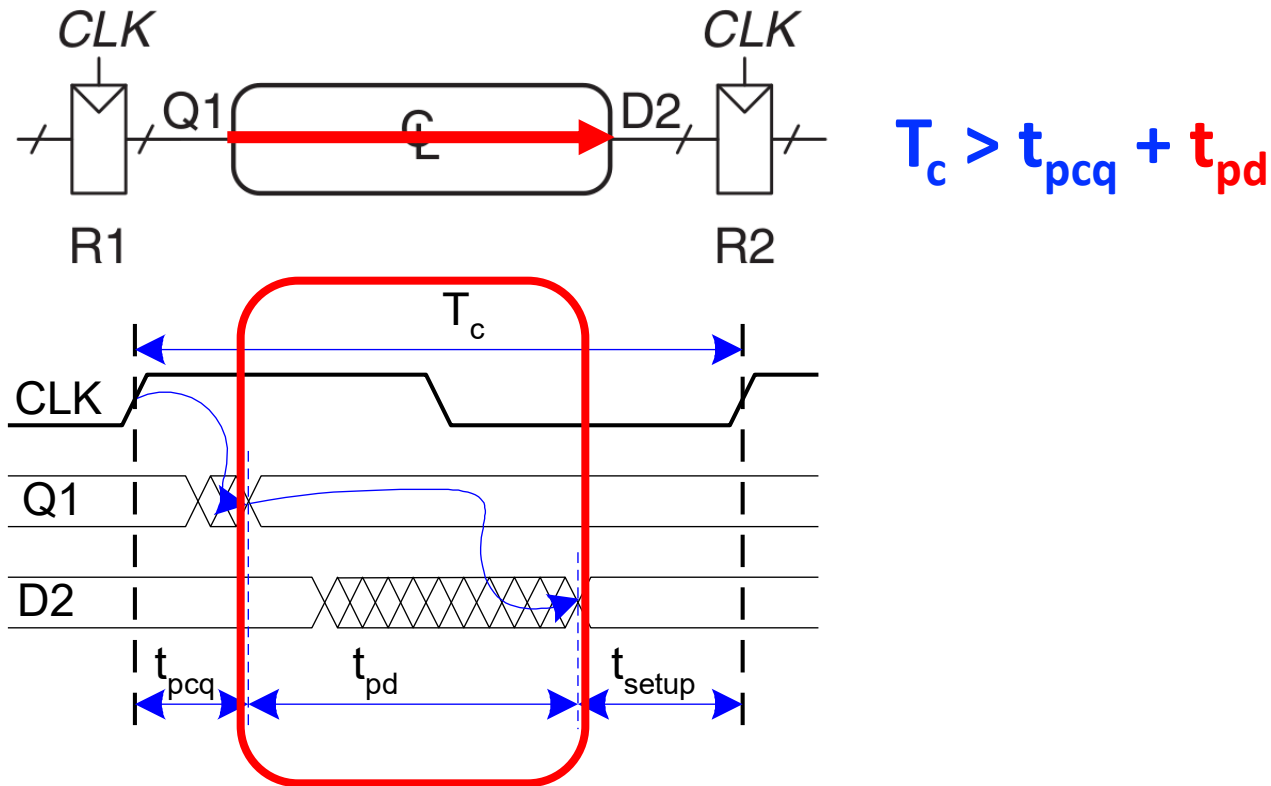
Setup Time Constraint

- **Safe timing** depends on the **maximum** delay from R1 to R2
- The input to R2 must be stable at least t_{setup} **before** the clock edge.



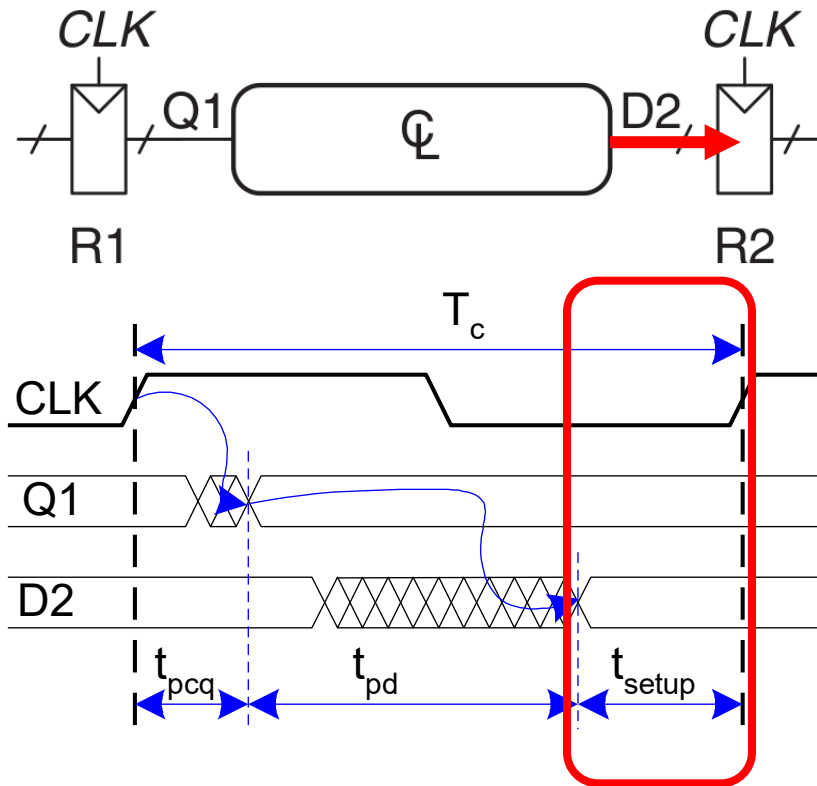
Setup Time Constraint

- **Safe timing** depends on the **maximum** delay from R1 to R2
- The input to R2 must be stable at least t_{setup} **before** the clock edge.



Setup Time Constraint

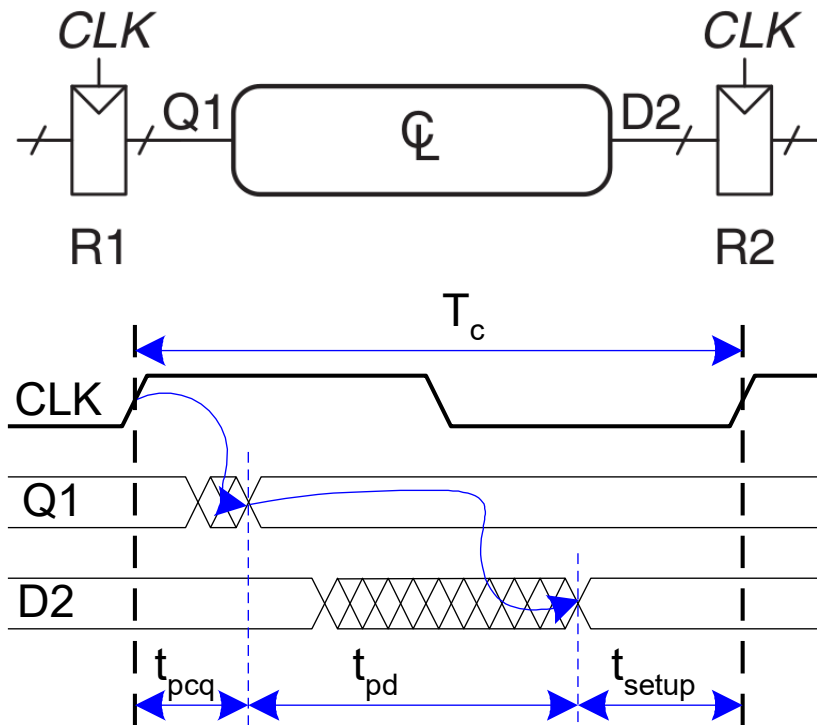
- **Safe timing** depends on the **maximum** delay from R1 to R2
- The input to R2 must be stable at least t_{setup} **before** the clock edge.



$$T_c > t_{\text{pcq}} + t_{\text{pd}} + t_{\text{setup}}$$

Setup Time Constraint

- **Safe timing** depends on the **maximum** delay from R1 to R2
- The input to R2 must be stable at least t_{setup} **before** the clock edge.

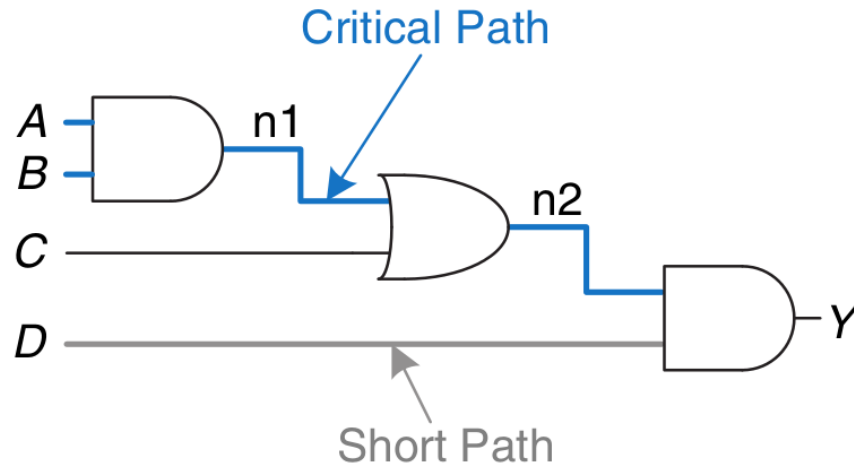


$$T_c > \underbrace{t_{\text{pcq}} + t_{\text{pd}}}_{\text{Useful work}} + t_{\text{setup}}$$

Wasted work

Sequencing overhead:
amount of time **wasted**
each cycle due to sequencing
element timing requirements

t_{setup} Constraint and Design Performance



- **Critical path:** path with the longest t_{pd}

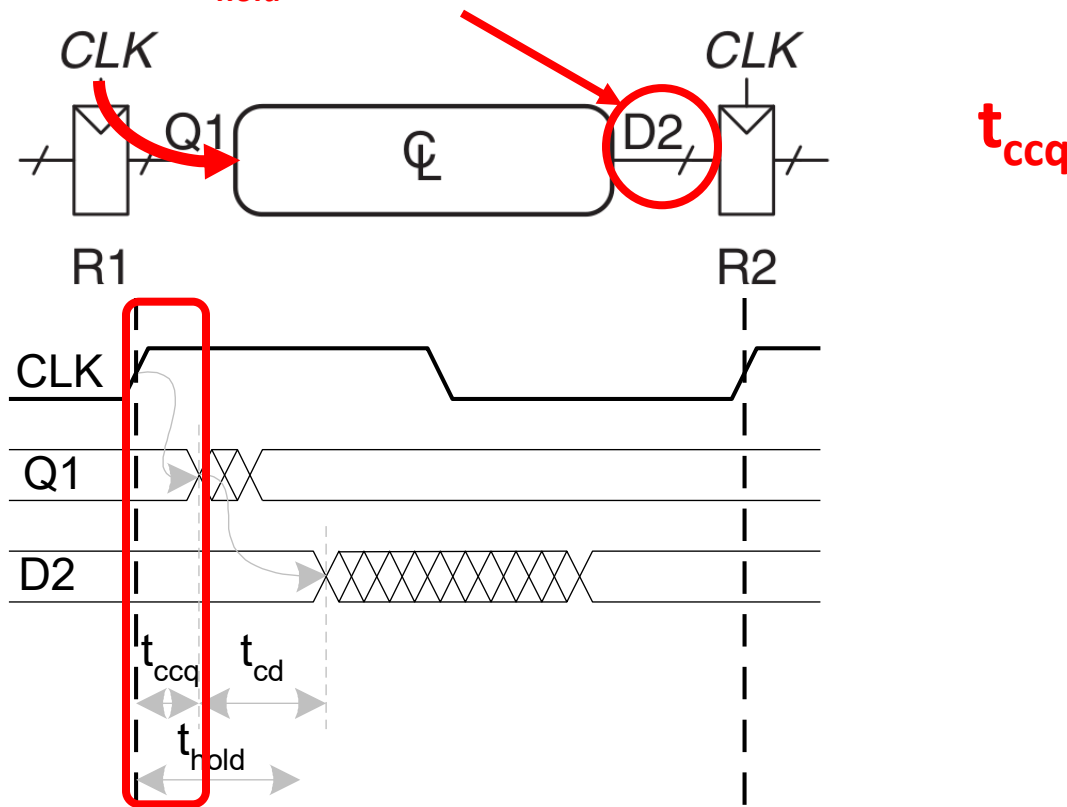
$$T_c > t_{\text{pcq}} + t_{\text{pd}} + t_{\text{setup}}$$

- Overall design performance is determined by the critical path t_{pd}
 - Determines the **minimum clock period** (i.e., **max operating frequency**)
 - If the critical path is too **long**, the design will run **slowly**
 - If critical path is too **short**, each cycle will do very **little useful work**
 - i.e., most of the cycle will be **wasted** in sequencing overhead

Hold Time Constraint

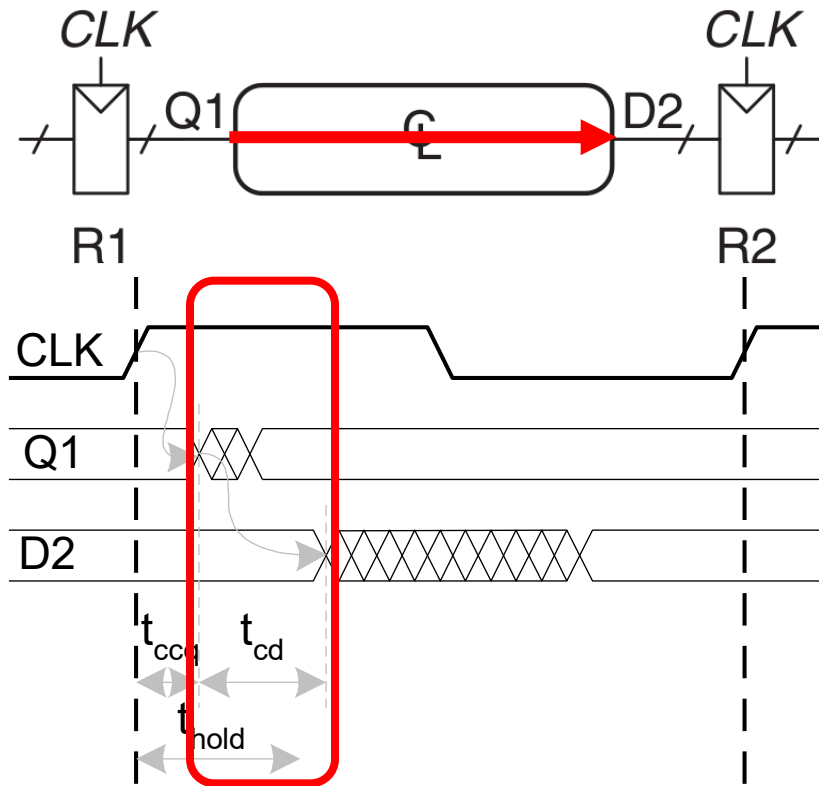
- **Safe timing** depends on the **minimum** delay from R1 to R2
- **D2** (i.e., R2 input) must be stable for at least t_{hold} **after** the clock edge

*Must not change until
 t_{hold} after the clock edge*



Hold Time Constraint

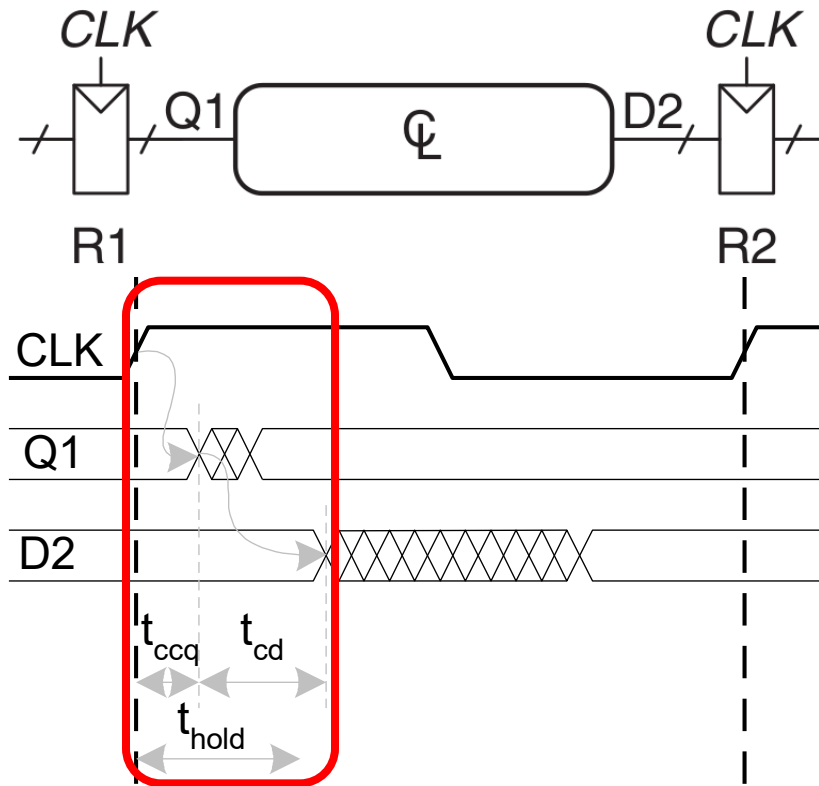
- **Safe timing** depends on the **minimum** delay from R1 to R2
- **D2** (i.e., R2 input) must be stable for at least t_{hold} **after** the clock edge



$$t_{\text{ccq}} + t_{\text{cd}}$$

Hold Time Constraint

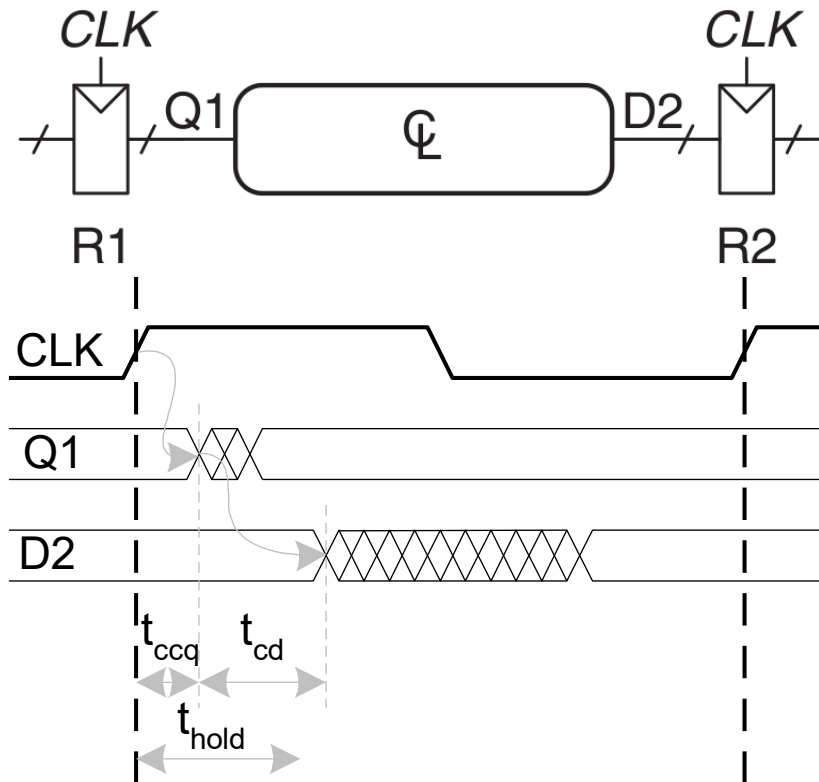
- **Safe timing** depends on the **minimum** delay from R1 to R2
- **D2** (i.e., R2 input) must be stable for at least t_{hold} **after** the clock edge



$$t_{\text{ccq}} + t_{\text{cd}} > t_{\text{hold}}$$

Hold Time Constraint

- **Safe timing** depends on the **minimum** delay from R1 to R2
- **D2** (i.e., R2 input) must be stable for at least t_{hold} **after** the clock edge



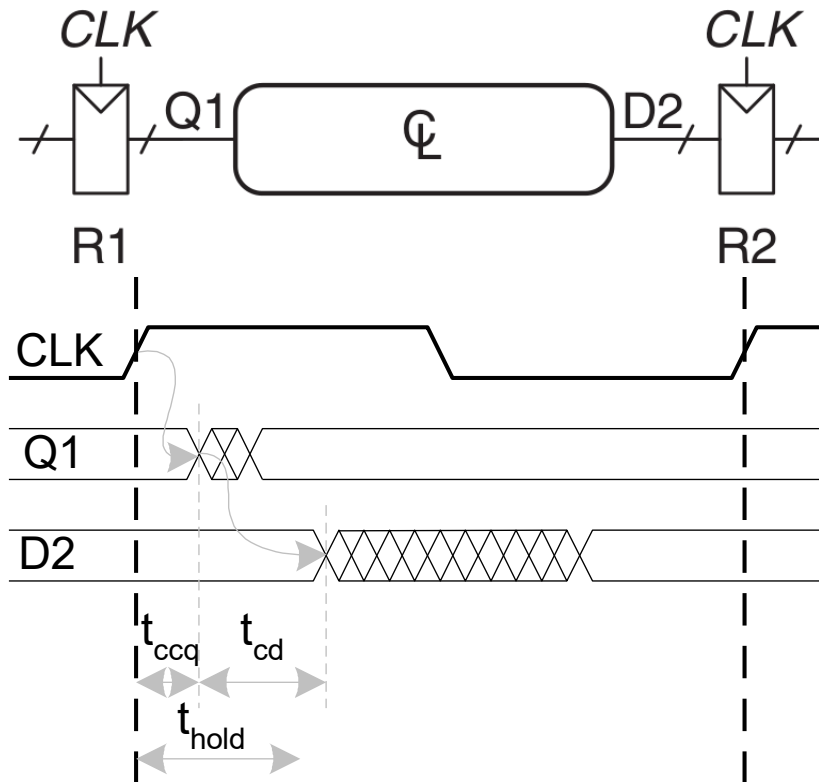
$$t_{\text{ccq}} + t_{\text{cd}} > t_{\text{hold}}$$

$$t_{\text{cd}} > t_{\text{hold}} - t_{\text{ccq}}$$

We need to have a **minimum** combinational delay!

Hold Time Constraint

- **Safe timing** depends on the **minimum** delay from R1 to R2
- **D2** (i.e., R2 input) must be stable for at least t_{hold} **after** the clock edge



$$t_{\text{ccq}} + t_{\text{cd}} > t_{\text{hold}}$$

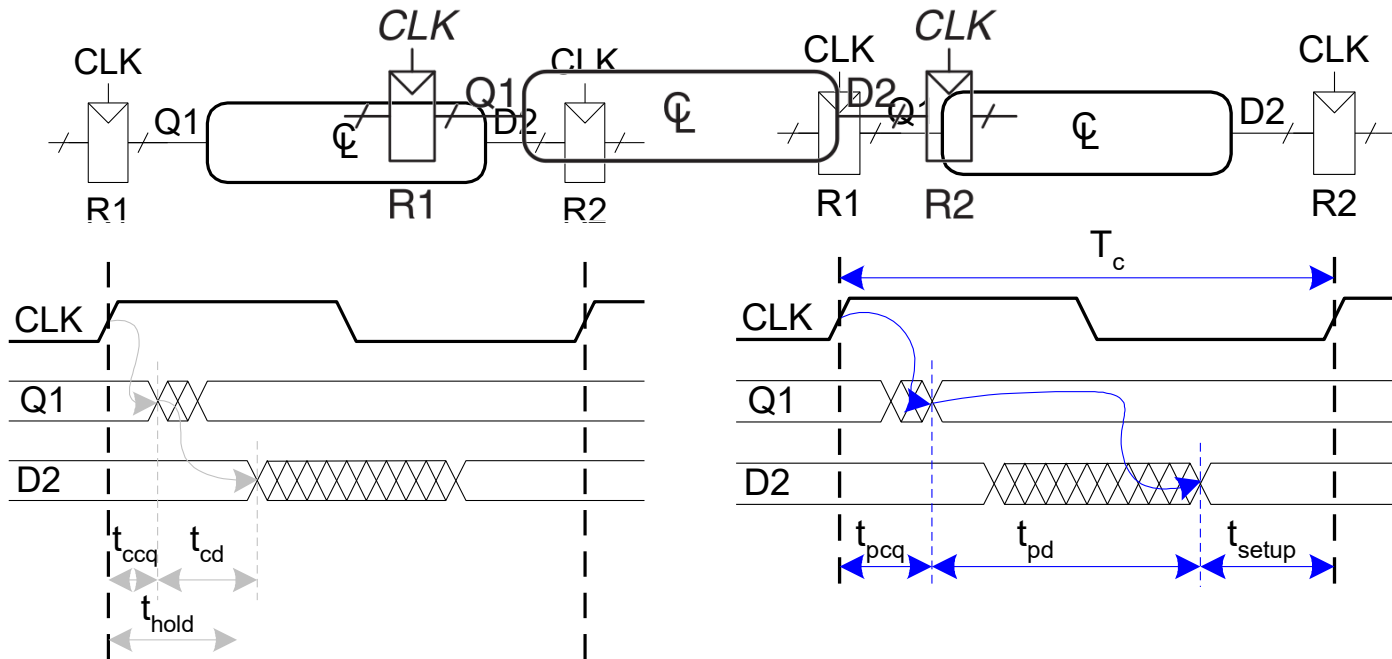
$$t_{\text{cd}} > t_{\text{hold}} - t_{\text{ccq}}$$

Does **NOT** depend on T_c !

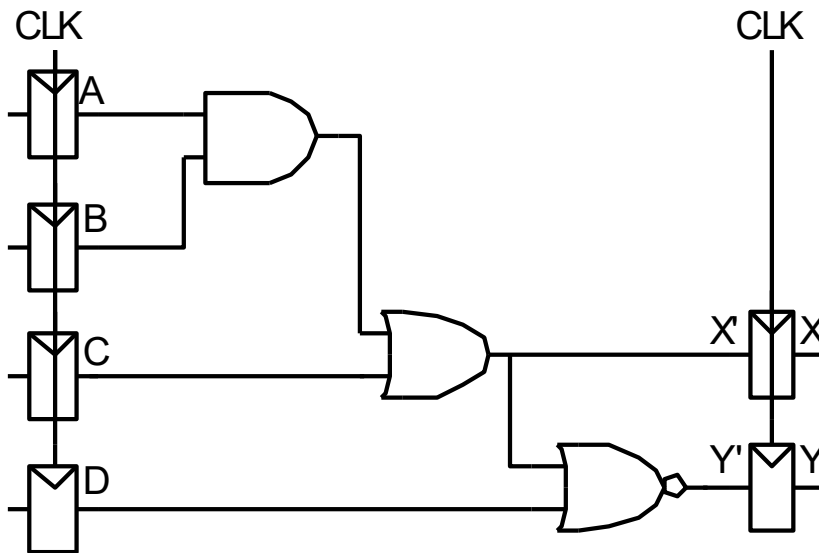
Very hard to fix t_{hold} violations after manufacturing- must modify circuits!

Sequential Timing Summary

t_{ccq} / t_{pcq}	clock-to-q delay (contamination/propagation)
t_{cd} / t_{pd}	combinational logic delay (contamination/propagation)
t_{setup}	time that FF inputs must be stable before next clock edge
t_{hold}	time that FF inputs must be stable after a clock edge
T_c	clock period



Example: Timing Analysis



Timing Characteristics

$$t_{ccq} = 30 \text{ ps}$$

$$t_{pcq} = 50 \text{ ps}$$

$$t_{\text{setup}} = 60 \text{ ps}$$

$$t_{\text{hold}} = 70 \text{ ps}$$

per gate

$$\begin{cases} t_{pd} = 35 \text{ ps} \\ t_{cd} = 25 \text{ ps} \end{cases}$$

$$t_{pd} =$$

$$t_{cd} =$$

Check setup time constraints:

$$T_c > t_{pcq} + t_{pd} + t_{\text{setup}}$$

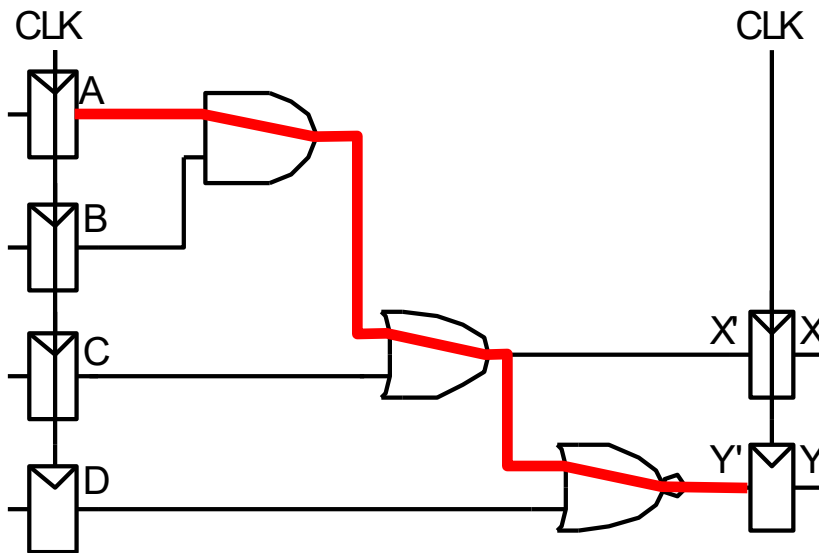
$$T_c >$$

$$f_{\text{max}} = 1/T_c =$$

Check hold time constraints:

$$t_{ccq} + t_{cd} > t_{\text{hold}} ?$$

Example: Timing Analysis



$$t_{pd} = 3 \times 35 \text{ ps} = 105 \text{ ps}$$

$$t_{cd} =$$

Check setup time constraints:

$$T_c > t_{pcq} + t_{pd} + t_{setup}$$

$$T_c >$$

$$f_{max} = 1/T_c =$$

Timing Characteristics

$$t_{ccq} = 30 \text{ ps}$$

$$t_{pcq} = 50 \text{ ps}$$

$$t_{setup} = 60 \text{ ps}$$

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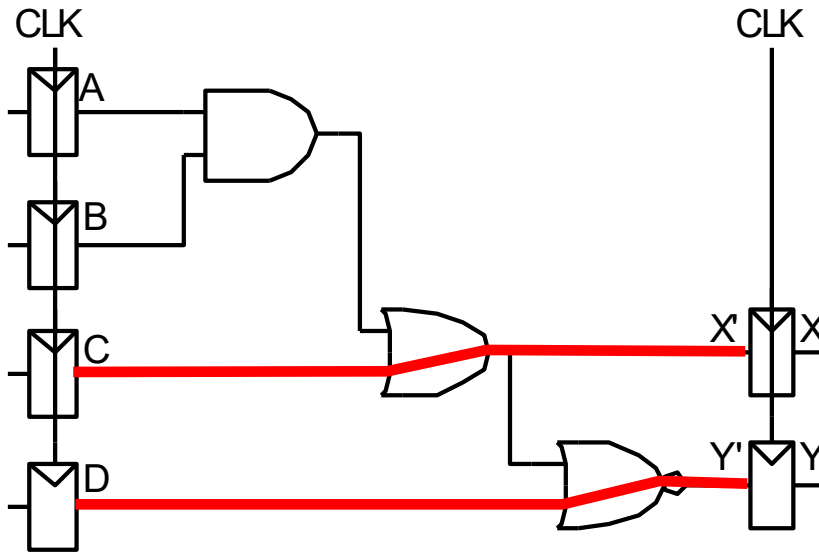
per gate

$$\begin{cases} t_{pd} = 35 \text{ ps} \\ t_{cd} = 25 \text{ ps} \end{cases}$$

Check hold time constraints:

$$t_{ccq} + t_{cd} > t_{hold} ?$$

Example: Timing Analysis



$$t_{pd} = 3 \times 35 \text{ ps} = 105 \text{ ps}$$

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Timing Characteristics

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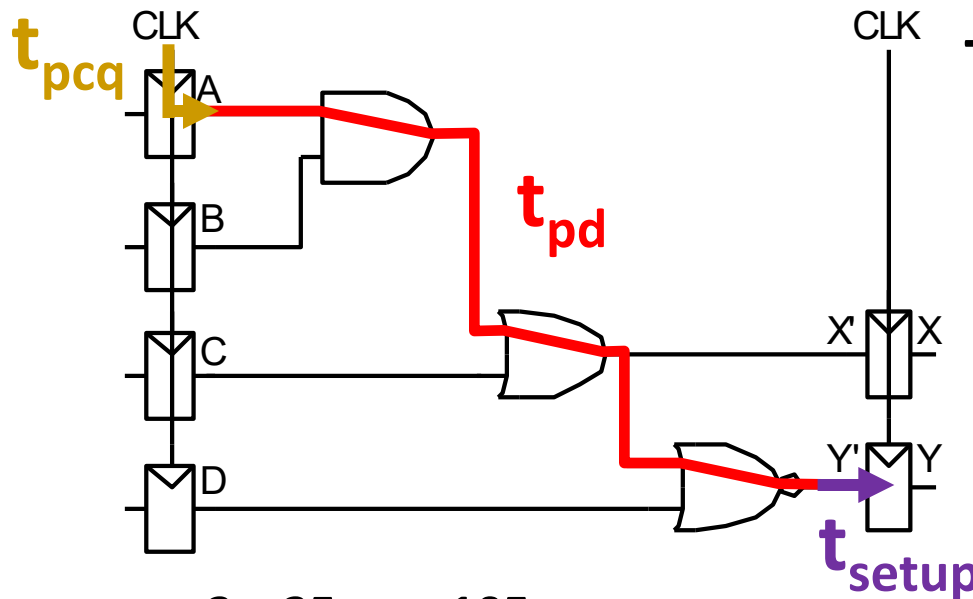
per gate

$$\left[\begin{array}{l} t_{pd} \\ t_{cd} \end{array} \right] = \left[\begin{array}{l} 35 \text{ ps} \\ 25 \text{ ps} \end{array} \right]$$

Check hold time constraints:

$$t_{ccq} + t_{cd} > t_{hold} ?$$

Example: Timing Analysis



$$t_{pd} = 3 \times 35 \text{ ps} = 105 \text{ ps}$$

$$t_{cd} = 25 \text{ ps}$$

Check setup time constraints:

$$T_c > t_{pcq} + t_{pd} + t_{setup}$$

$$T_c > (50 + 105 + 60) \text{ ps} = 215 \text{ ps}$$

$$f_{max} = 1/T_c = 4.65 \text{ GHz}$$

Timing Characteristics

$$t_{ccq} = 30 \text{ ps}$$

$$t_{pcq} = 50 \text{ ps}$$

$$t_{setup} = 60 \text{ ps}$$

$$t_{hold} = 70 \text{ ps}$$

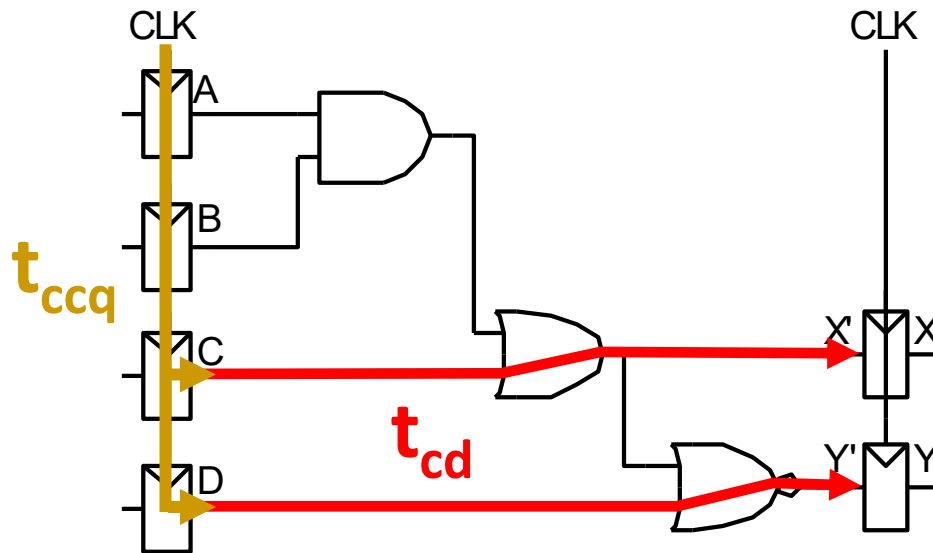
per gate

$$\begin{cases} t_{pd} = 35 \text{ ps} \\ t_{cd} = 25 \text{ ps} \end{cases}$$

Check hold time constraints:

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Example: Timing Analysis



$$t_{pd} = 3 \times 35 \text{ ps} = 105 \text{ ps}$$

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Check setup time constraints:

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Timing Characteristics

$$t_{ccq} = 30 \text{ ps}$$

$$t_{pcq} = 50 \text{ ps}$$

$$t_{setup} = 60 \text{ ps}$$

$$t_{hold} = 70 \text{ ps}$$

per gate

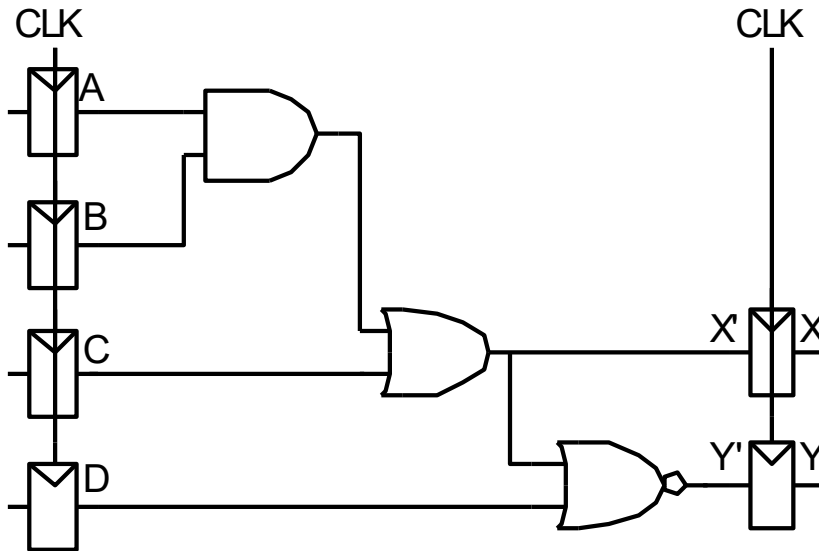
$$\begin{cases} t_{pd} = 35 \text{ ps} \\ t_{cd} = 25 \text{ ps} \end{cases}$$

Check hold time constraints:

$$t_{ccq} + t_{cd} > t_{hold} ?$$

$$(30 + 25) \text{ ps} > 70 \text{ ps} ?$$

Example: Timing Analysis



$$t_{pd} = 3 \times 35 \text{ ps} = 105 \text{ ps}$$

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Check setup time constraints:

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Timing Characteristics

$$t_{ccq} = 30 \text{ ps}$$

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per gate

$$\left[\begin{array}{l} t_{pd} = 35 \text{ ps} \\ t_{cd} = 25 \text{ ps} \end{array} \right]$$

Check hold time constraints:

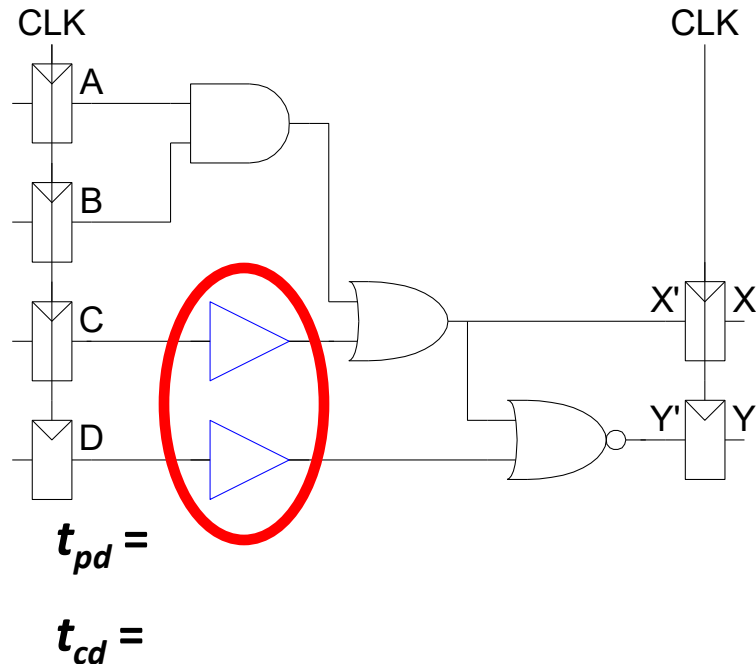
$$t_{ccq} + t_{cd} > t_{hold} ?$$

$$(30 + 25) \text{ ps} > 70 \text{ ps} ?$$

FAIL

Example: Fixing Hold Time Violation

Add buffers to the short paths:



Check setup time constraints:

$$T_c > t_{pcq} + t_{pd} + t_{setup}$$

$$T_c >$$

$$f_c =$$

Timing Characteristics

$$t_{ccq} = 30 \text{ ps}$$

$$t_{pcq} = 50 \text{ ps}$$

$$t_{setup} = 60 \text{ ps}$$

$$t_{hold} = 70 \text{ ps}$$

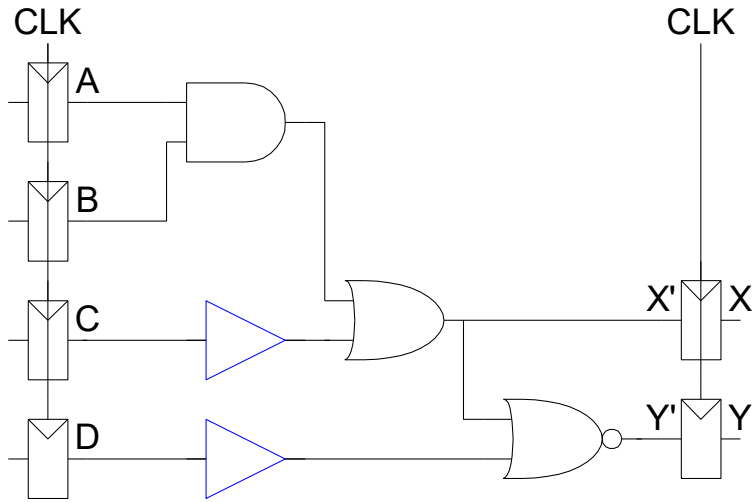
$$\text{per gate} \left[\begin{array}{l} t_{pd} = 35 \text{ ps} \\ t_{cd} = 25 \text{ ps} \end{array} \right]$$

Check hold time constraints:

$$t_{ccq} + t_{cd} > t_{hold} ?$$

Example: Fixing Hold Time Violation

Add buffers to the short paths:



$$t_{pd} = 3 \times 35 \text{ ps} = 105 \text{ ps}$$

$$t_{cd} = 2 \times 25 \text{ ps} = 50 \text{ ps}$$

Check setup time constraints:

$$T_c > t_{pcq} + t_{pd} + t_{setup}$$

$T_c >$

$$f_c =$$

Timing Characteristics

$$t_{ccq} = 30 \text{ ps}$$

$$t_{pcq} = 50 \text{ ps}$$

$$t_{\text{setup}} = 60 \text{ ps}$$

$$t_{\text{hold}} = 70 \text{ ps}$$

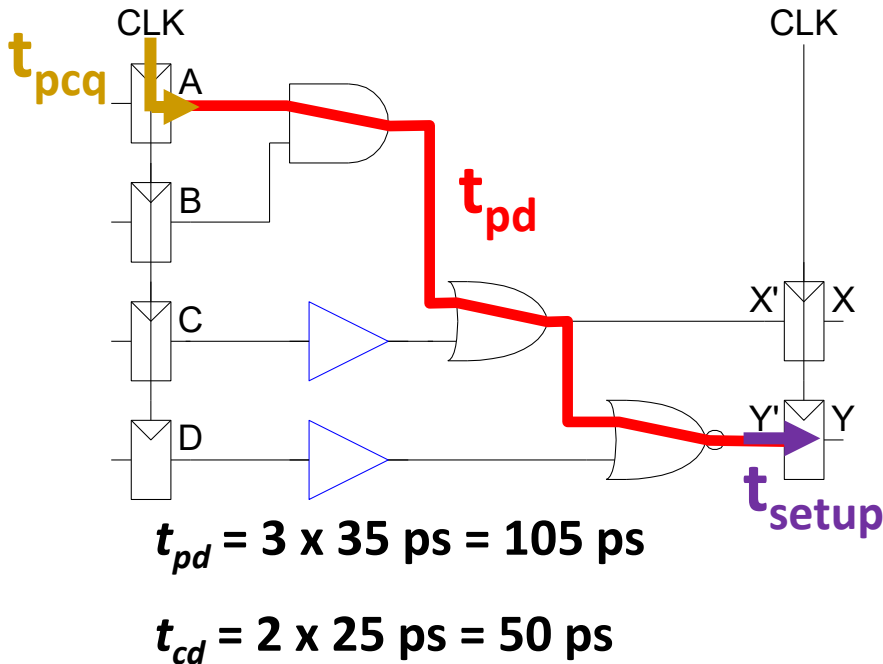
per gate $\left[\begin{array}{ll} t_{pd} & = 35 \text{ ps} \\ t_{cd} & = 25 \text{ ps} \end{array} \right.$

Check hold time constraints:

$$t_{ccq} + t_{cd} > t_{hold} ?$$

Example: Fixing Hold Time Violation

Add buffers to the short paths:



Check setup time constraints:

$$T_c > t_{pcq} + t_{pd} + t_{setup}$$

$$T_c > (50 + 105 + 60) \text{ ps} = 215 \text{ ps}$$

$$f_c = 1/T_c = 4.65 \text{ GHz}$$

Timing Characteristics

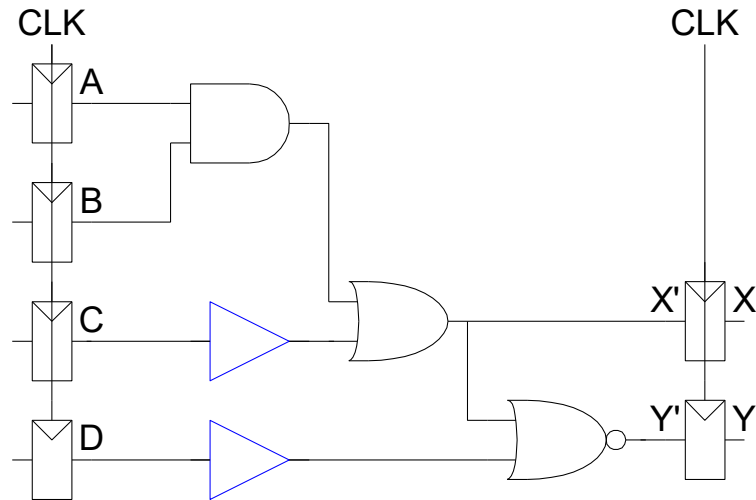
t_{ccq}	= 30 ps
t_{pcq}	= 50 ps
t_{setup}	= 60 ps
t_{hold}	= 70 ps
per gate	t_{pd} = 35 ps
	t_{cd} = 25 ps

Check hold time constraints:

$$t_{ccq} + t_{cd} > t_{hold} ?$$

Example: Fixing Hold Time Violation

Add buffers to the short paths:



$$t_{pd} = 3 \times 35 \text{ ps} = 105 \text{ ps}$$

$$t_{cd} = 2 \times 25 \text{ ps} = 50 \text{ ps}$$

Check setup time constraints:

$$T_c > t_{pcq} + t_{pd} + t_{setup}$$

$$T_c > (50 + 105 + 60) \text{ ps} = 215 \text{ ps}$$

$$f_c = 1/T_c = \mathbf{4.65 \text{ GHz}}$$

Note: no change to max frequency!

Timing Characteristics

$$t_{ccq} = 30 \text{ ps}$$

$$t_{pcq} = 50 \text{ ps}$$

$$t_{setup} = 60 \text{ ps}$$

$$t_{hold} = 70 \text{ ps}$$

per gate

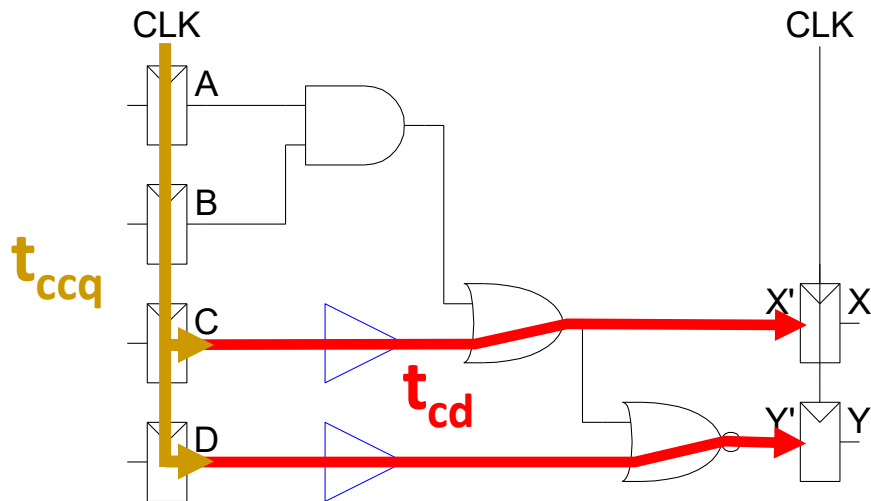
$$\left[\begin{array}{l} t_{pd} \\ t_{cd} \end{array} \right] = \left[\begin{array}{l} 35 \text{ ps} \\ 25 \text{ ps} \end{array} \right]$$

Check hold time constraints:

$$t_{ccq} + t_{cd} > t_{hold} ?$$

Example: Fixing Hold Time Violation

Add buffers to the short paths:



$$t_{pd} = 3 \times 35 \text{ ps} = 105 \text{ ps}$$

$$t_{cd} = 2 \times 25 \text{ ps} = 50 \text{ ps}$$

Check setup time constraints:

$$T_c > t_{pcq} + t_{pd} + t_{setup}$$

$$T_c > (50 + 105 + 60) \text{ ps} = 215 \text{ ps}$$

$$f_c = 1/T_c = 4.65 \text{ GHz}$$

Timing Characteristics

$$t_{ccq} = 30 \text{ ps}$$

$$t_{pcq} = 50 \text{ ps}$$

$$t_{setup} = 60 \text{ ps}$$

$$t_{hold} = 70 \text{ ps}$$

$$\text{per gate} \left[\begin{array}{l} t_{pd} = 35 \text{ ps} \\ t_{cd} = 25 \text{ ps} \end{array} \right]$$

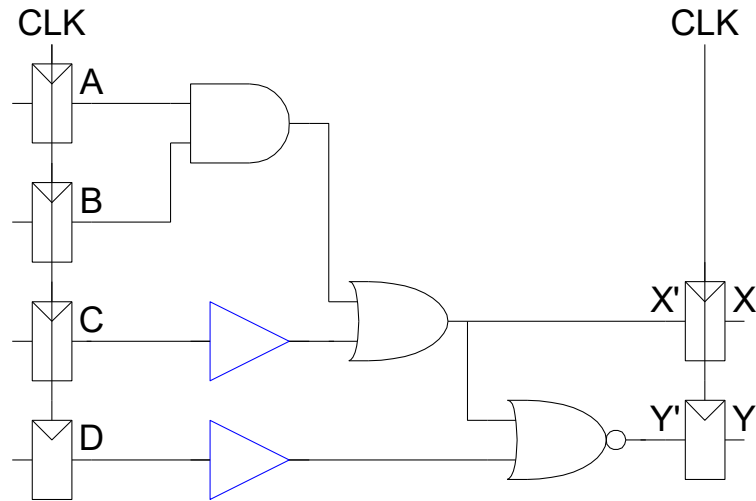
Check hold time constraints:

$$t_{ccq} + t_{cd} > t_{hold} ?$$

$$(30 + 50) \text{ ps} > 70 \text{ ps} ?$$

Example: Fixing Hold Time Violation

Add buffers to the short paths:



$$t_{pd} = 3 \times 35 \text{ ps} = 105 \text{ ps}$$

$$t_{cd} = 2 \times 25 \text{ ps} = 50 \text{ ps}$$

Check setup time constraints:

$$T_c > t_{pcq} + t_{pd} + t_{setup}$$

$$T_c > (50 + 105 + 60) \text{ ps} = 215 \text{ ps}$$

$$f_c = 1/T_c = 4.65 \text{ GHz}$$

Timing Characteristics

$$t_{ccq} = 30 \text{ ps}$$

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$$t_{setup} = 60 \text{ ps}$$

$$t_{hold} = 70 \text{ ps}$$

$$\text{per gate} \left[\begin{array}{l} t_{pd} = 35 \text{ ps} \\ t_{cd} = 25 \text{ ps} \end{array} \right]$$

Check hold time constraints:

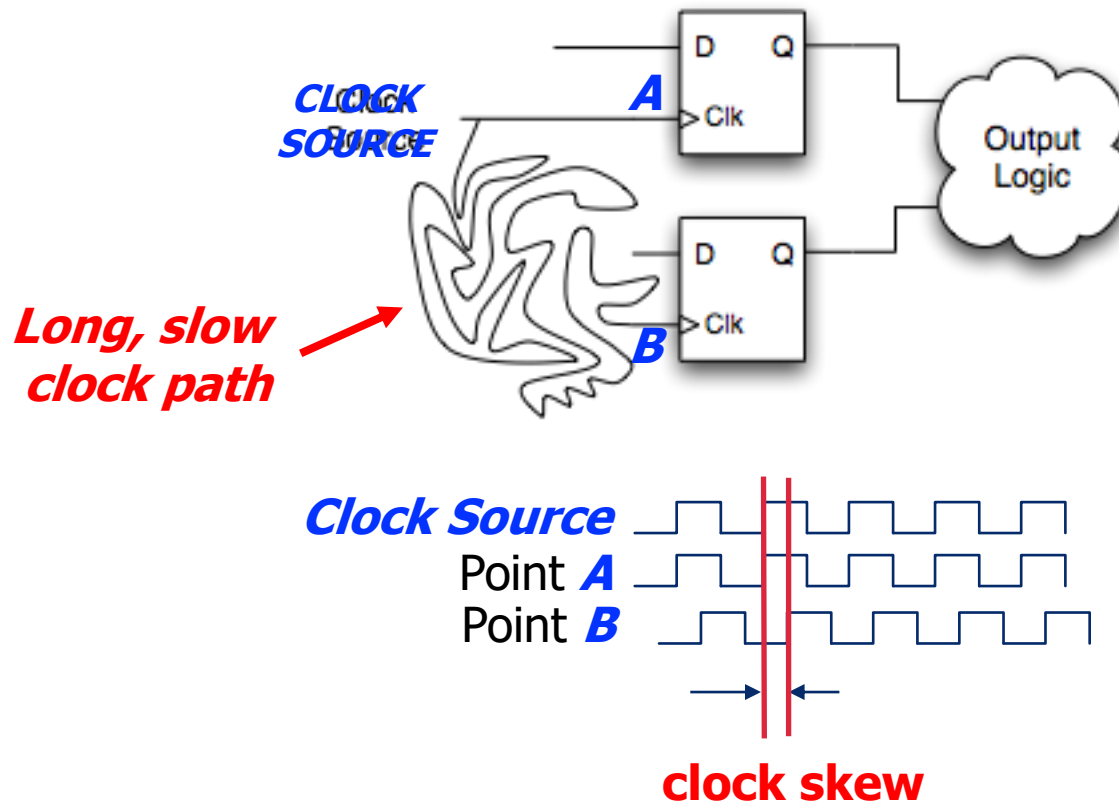
$$t_{ccq} + t_{cd} > t_{hold} ?$$

$$(30 + 50) \text{ ps} > 70 \text{ ps} ?$$

PASS

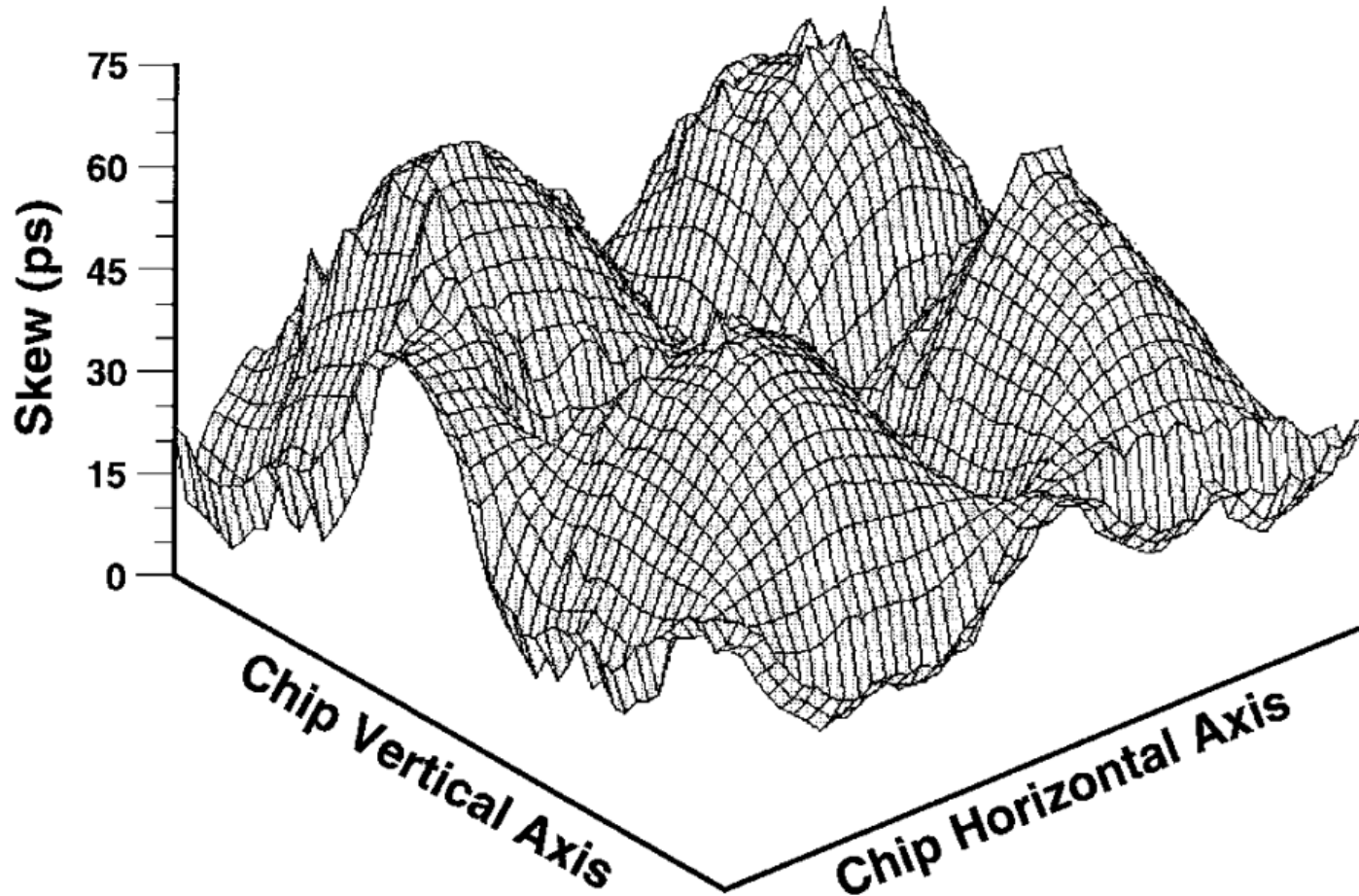
Clock Skew

- To make matters worse, **clocks have delay** too!
 - The clock does **not** reach all parts of the chip at the same time!
- **Clock skew**: time difference between two clock edges



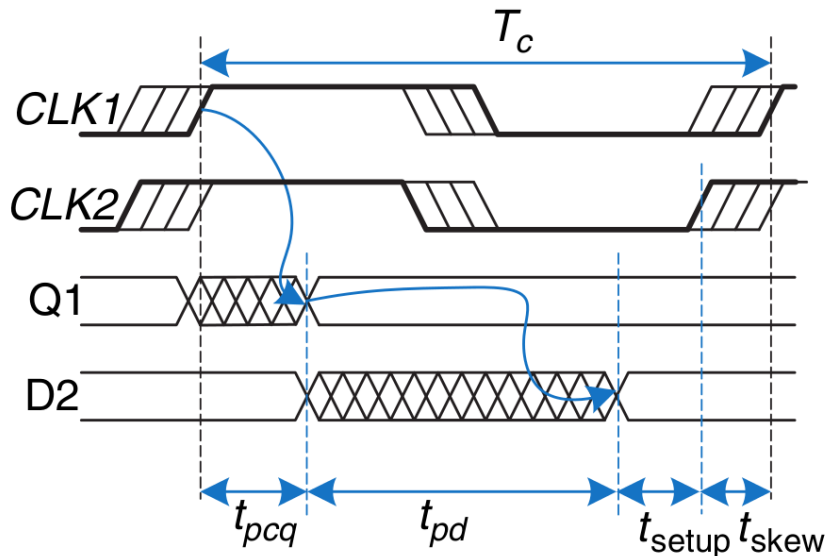
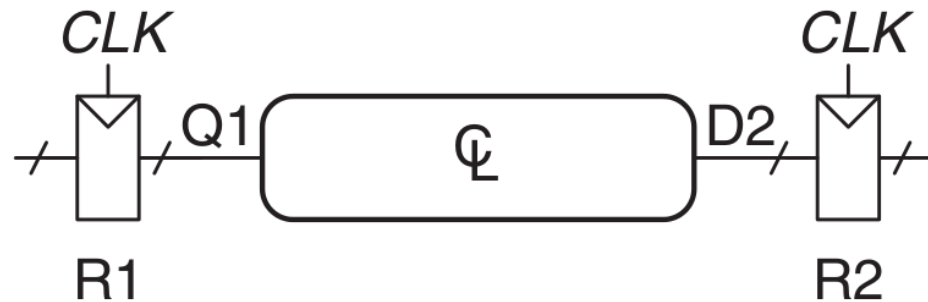
Clock Skew Example

- Example of the **Alpha 21264** clock skew spatial distribution



Clock Skew: Setup Time Revisited

- **Safe timing** requires considering the **worst-case skew**
 - Clock arrives at **R2 before R1**
 - Leaves **as little time as possible** for the **combinational logic**



Signal must arrive at D2 **earlier!**

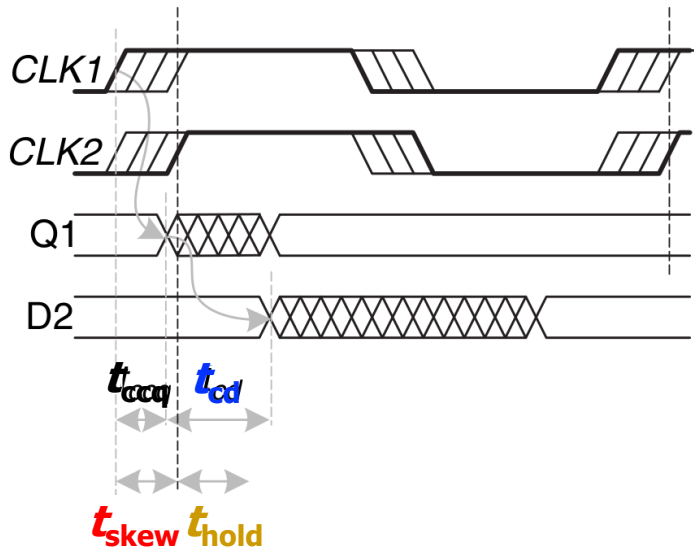
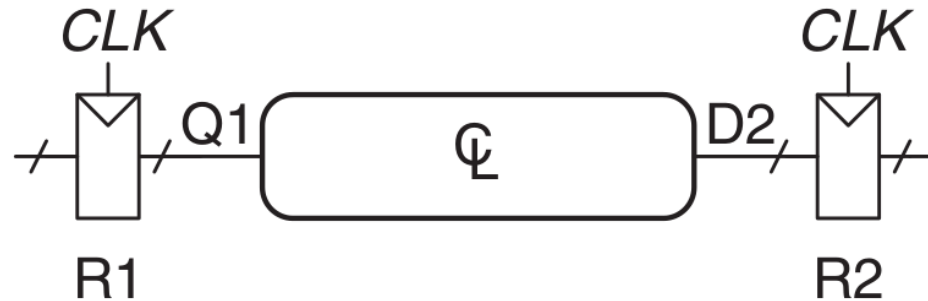
This effectively **increases** t_{setup} :

$$T_c > t_{pcq} + t_{pd} + t_{setup} + t_{skew}$$

$$T_c > t_{pcq} + t_{pd} + t_{setup, effective}$$

Clock Skew: Hold Time Revisited

- **Safe timing** requires considering the **worst-case skew**
 - Clock arrives at **R2** *after* **R1**
 - Increases the **minimum required delay** for the **combinational logic**



Signal must arrive at D2 *later*!

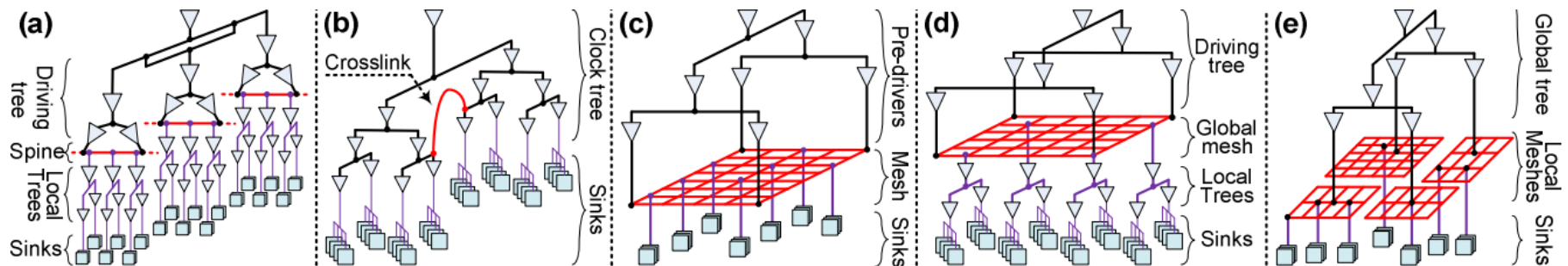
This effectively *increases* t_{hold} :

$$t_{\text{cd}} + t_{\text{ccq}} > \underbrace{t_{\text{hold}} + t_{\text{skew}}}$$

$$t_{\text{cd}} + t_{\text{ccq}} > t_{\text{hold, effective}}$$

Clock Skew: Summary

- **Skew** effectively **increases** both t_{setup} and t_{hold}
 - Increased **sequencing overhead**
 - i.e., less useful work done per cycle
- Designers must keep skew to a **minimum**
 - Requires intelligent **"clock network"** across a chip
 - **Goal: clock** arrives at all locations at roughly the **same time**



Source: Abdelhadi, Ameer, et al. "Timing-driven variation-aware nonuniform clock mesh synthesis." GLSVLSI'10.

Part 3:

Circuit Verification

How Do You Know That A Circuit Works?

- You have designed a circuit
 - Is it **functionally** correct?
 - Even if it is logically correct, does the hardware meet all **timing** constraints?
- How can you **test** for:
 - Functionality?
 - Timing?
- Answer: **simulation tools!**
 - Formal verification tools (e.g., SAT solvers)
 - HDL timing simulation (e.g., Vivado)
 - Circuit simulation (e.g., SPICE)

Testing Large Digital Designs

- Testing can be the **most time consuming** design stage
 - Functional correctness of **all logic paths**
 - Timing, power, etc. of **all circuit elements**

- Unfortunately, **low-level** (e.g., circuit) simulation is **much slower** than **high-level** (e.g., HDL, C) simulation

- **Solution:** we split responsibilities:
 - 1) Check **only functionality** at a **high level** (e.g., C, HDL)
 - (Relatively) **fast** simulation time allows **high code coverage**
 - **Easy** to write and run tests
 - 2) Check **only timing, power**, etc. at **low level** (e.g., circuit)
 - **No functional testing** of low-level model
 - Instead, test **functional equivalence** to high-level model
 - **Hard**, but **easier** than testing logical functionality at this level

Testing Large Digital Designs

- We have **tools** to handle different levels of verification
 - *Logic synthesis tools* guarantee equivalence of high-level logic and synthesized circuit-level description
 - *Timing verification tools* check all **circuit timings**
 - *Design rule checks* ensure that **physical circuits** are buildable
- **The task of a logic designer** is to:
 - Provide **functional tests** for logical correctness of the design
 - Provide **timing constraints** (e.g., desired operating frequency)
- Tools and/or circuit engineers will decide if it can be built!

Part 4:

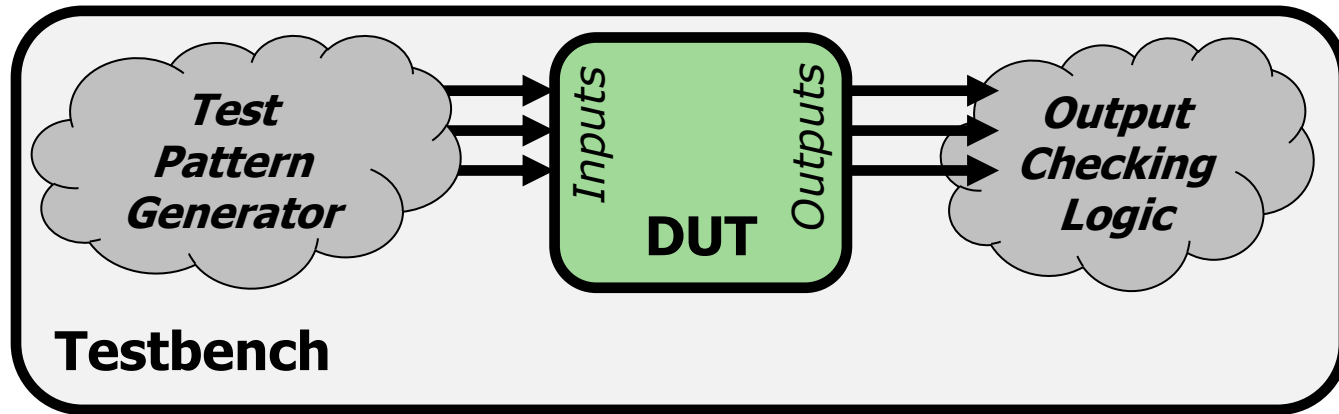
Functional Verification

Functional Verification

- Goal: check **logical correctness** of the design
- Physical circuit timing (e.g., $t_{\text{setup}}/t_{\text{hold}}$) is typically ignored
 - May implement simple checks to catch obvious bugs
 - We'll discuss timing verification later in this lecture
- There are two primary approaches
 - Logic simulation (e.g., C/C++/Verilog test routines)
 - Formal verification techniques
- In this course, we will use Verilog for functional verification

Testbench-Based Functional Testing

- **Testbench:** a module created specifically to test a design
 - Tested design is called the “**device under test (DUT)**”



- Testbench **provides inputs (test patterns)** to the DUT
 - Hand-crafted values
 - Automatically generated (e.g., sequential or random values)
- Testbench **checks outputs** of the DUT against:
 - Hand-crafted values
 - A “golden design” that is known to be bug-free

Testbench-Based Functional Testing

- A testbench can be:
 - **HDL code** written to test other HDL modules
 - **Circuit schematic** used to test other circuit designs
- The testbench is **not** designed for **hardware synthesis!**
 - Runs in **simulation** only
 - HDL simulator (e.g., Vivado simulator)
 - SPICE circuit simulation
 - Testbench uses **simulation-only** constructs
 - E.g., “wait 10ns”
 - E.g., ideal voltage/current source
 - Not suitable to be physically built!

Common Verilog Testbench Types

Testbench	Input/Output Generation	Error Checking
Simple	Manual	Manual
Self-Checking	Manual	Automatic
Automatic	Automatic	Automatic

Example DUT

- We will walk through different types of testbenches to test a module that implements the logic function:

$$y = (\bar{b} \cdot \bar{c}) + (a \cdot \bar{b})$$

```
// performs y = ~b & ~c | a & ~b
module sillyfunction(input  a, b, c,
                    output y);

    wire b_n, c_n;
    wire m1, m2;

    not not_b(b_n, b);
    not not_c(c_n, c);

    and minterm1(m1, b_n, c_n);
    and minterm2(m2, a, b_n);
    or  out_func(y, m1, m2);

endmodule
```

Useful Verilog Syntax for Testbenching

```
module example_syntax();  
    reg a;  
  
    // like "always" block, but runs only once at sim start  
    initial  
    begin  
        a = 0; // set value of reg: use blocking assignments  
        #10;    // wait (do nothing) for 10 ns  
        a = 1;  
        $display("printf() style message!"); // print message  
    end  
endmodule
```

Simple Testbench

Simple Testbench

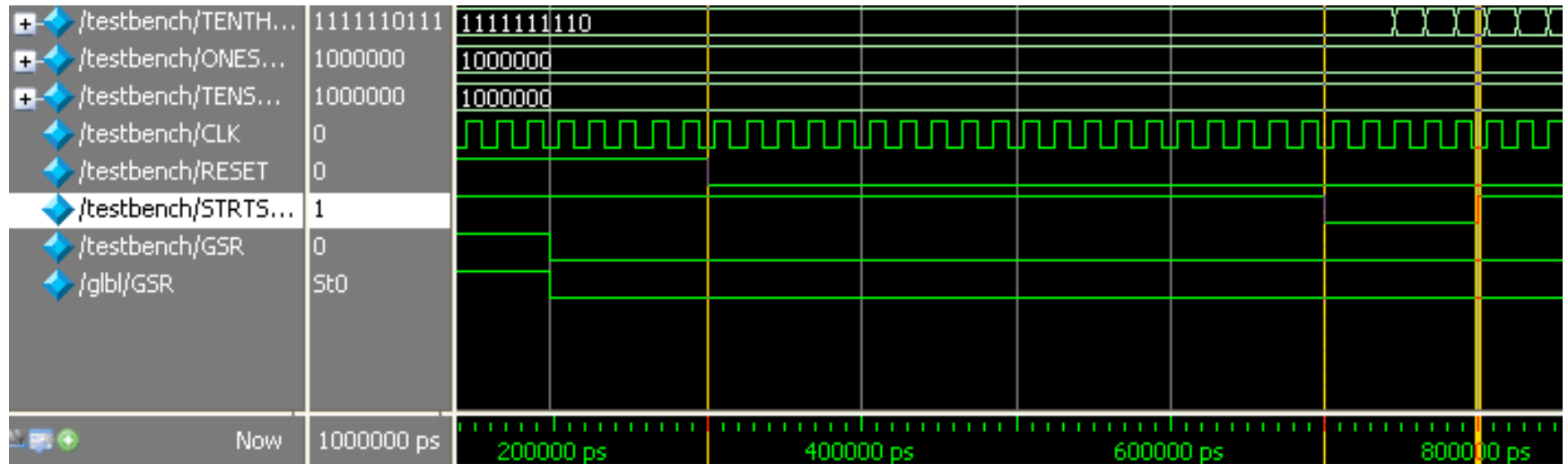
```
module testbench1(); // No inputs, outputs
    reg    a, b, c;      // Manually assigned
    wire  y;            // Manually checked

    // instantiate device under test
    sillyfunction dut (.a(a), .b(b), .c(c), .y(y) );

    // apply hardcoded inputs one at a time
    initial begin
        a = 0; b = 0; c = 0; #10; // apply inputs, wait 10ns
        c = 1; #10;                // apply inputs, wait 10ns
        b = 1; c = 0; #10;         // etc .. etc..
        c = 1; #10;
        a = 1; b = 0; c = 0; #10;
    end
endmodule
```

Simple Testbench: Output Checking

- Most common method is to look at **waveform diagrams**
 - ❑ ***Thousands*** of signals over ***millions*** of clock cycles
 - ❑ Too many to just printf()!



time →

- **Manually check** that output is correct **at all times**

Simple Testbench

■ **Pros:**

- ❑ Easy to design
- ❑ Can easily test a few, specific inputs (e.g., corner cases)

■ **Cons:**

- ❑ **Not scalable** to many test cases
- ❑ Outputs must be checked **manually** outside of the simulation
 - E.g., inspecting dumped waveform signals
 - E.g., printf() style debugging

Self-Checking Testbench

Self-Checking Testbench

```
module testbench2();  
    reg a, b, c;  
    wire y;  
  
    sillyfunction dut(.a(a), .b(b), .c(c), .y(y));  
  
    initial begin  
        a = 0; b = 0; c = 0; #10; // apply input, wait 10ns  
        if (y !== 1) $display("000 failed."); // check result  
        c = 1; #10;  
        if (y !== 0) $display("001 failed.");  
        b = 1; c = 0; #10;  
        if (y !== 0) $display("010 failed.");  
    end  
endmodule
```

Self-Checking Testbench

■ Pros:

- ❑ Still easy to design
- ❑ Still easy to test a few, specific inputs (e.g., corner cases)
- ❑ **Simulator will print** whenever an error occurs

■ Cons:

- ❑ **Still not scalable** to millions of test cases
- ❑ Easy to make an **error** in **hardcoded** values
 - You make just as many **errors** writing a testbench as actual code
 - **Hard to debug** whether an issue is in the testbench or in the DUT

Self-Checking Testbench using Testvectors

- Write *testvector file*
 - List of inputs and expected outputs
 - Can create vectors **manually** *or* **automatically** using an already verified, simpler **"golden model"** (more on this later)
- Example file:

```
$ cat testvectors.tv
```

```
000_1
```

```
001_0
```

```
010_0
```

```
011_0
```

```
100_1
```

```
101_1
```

```
110_0
```

```
111_0
```

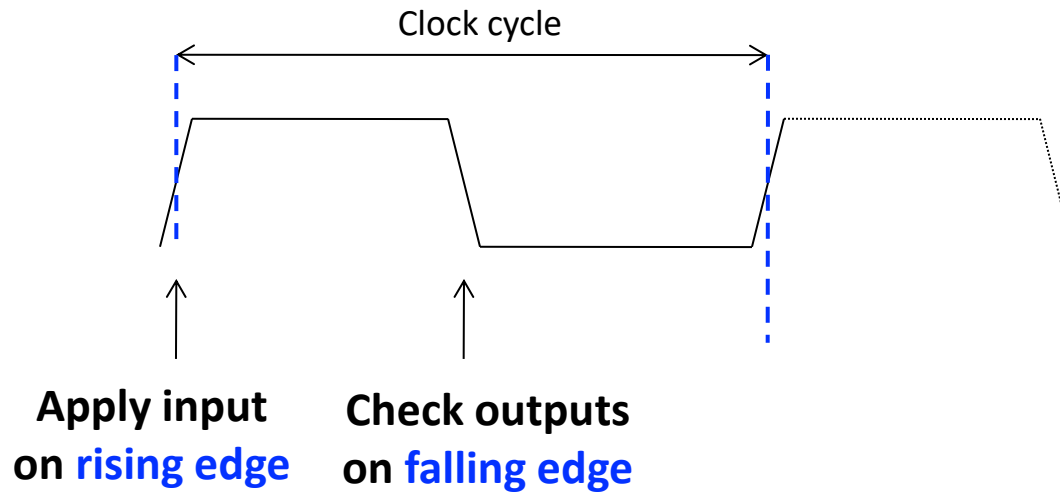
```
...
```

Format:
input_output



Testbench with Testvectors Design

- Use a “**clock signal**” for assigning inputs, reading outputs
 - Test one **testvector** each “**clock cycle**”



- Note: “clock signal” simply separates **inputs** from **outputs**
 - Allows us to *observe* the inputs/outputs in waveform diagrams
 - Not used for checking physical circuit timing (e.g., t_{setup} / t_{hold})
 - We’ll discuss *circuit timing verification* later in this lecture

Testbench Example (1 / 5): Signal Declarations

- Declare signals to hold internal state

```
module testbench3();  
    reg          clk, reset;      // clock and reset are internal  
    reg          a, b, c, yexpected; // values from testvectors  
    wire         y;                // output of circuit  
    reg [31:0] vectornum, errors;  // bookkeeping variables  
    reg [3:0]  testvectors[10000:0]; // array of testvectors  
  
    // instantiate device under test  
    sillyfunction dut(.a(a), .b(b), .c(c), .y(y) );
```

H&H Section 4.9, Example 4.39

Testbench Example (2/5): Clock Generation

```
// generate clock
always          // no sensitivity list, so it always executes
begin
    clk = 1; #5; clk = 0; #5;          // 10ns period
end
```


Testbench Example (3/5): Read Testvectors into Array

```
// at start of test, load vectors and pulse reset
initial    // Only executes once
begin
    $readmemb("example.tv", testvectors); // Read vectors
    vectornum = 0; errors = 0;              // Initialize
    reset = 1; #27; reset = 0;             // Apply reset wait
end

// Note: $readmemb reads testvector files written in
// hexadecimal
```

Testbench Example (4/5): Assign Inputs/Outputs

```
// apply test vectors on rising edge of clk
always @(posedge clk)
begin
    {a, b, c, yexpected} = testvectors[vectornum];
end
```

- Apply {a, b, c} inputs on the *rising edge* of the clock
- Get **yexpected** for checking the output on the *falling edge*
- Rising/falling edges are chosen only by convention
 - You can use any part of the clock signal
 - Your H+H textbook uses this convention

Testbench Example (5/5): Check Outputs

```
always @(negedge clk)
begin
    if (~reset) // don't test during reset
    begin
        if (y !== yexpected)
        begin
            $display("Error: inputs = %b", {a, b, c});
            $display("  outputs = %b (%b exp)", y, yexpected);
            errors = errors + 1;
        end

        // increment array index and read next testvector
        vectornum = vectornum + 1;

        if (testvectors[vectornum] === 4'bx)
        begin
            $display("%d tests completed with %d errors",
                    vectornum, errors);
            $finish;                // End simulation
        end
    end
end
end
```

Self-Checking Testbench with Testvectors

■ Pros:

- ❑ Still easy to design
- ❑ Still easy to test a few, specific inputs (e.g., corner cases)
- ❑ Simulator will print whenever an error occurs
- ❑ **No need** to change hardcoded values for **different tests**

■ Cons:

- ❑ May be **error-prone** depending on source of testvectors
- ❑ More scalable, but still **limited** by reading a file
 - Might have many more combinational paths to test than will fit in memory

Automatic Testbench

Golden Models

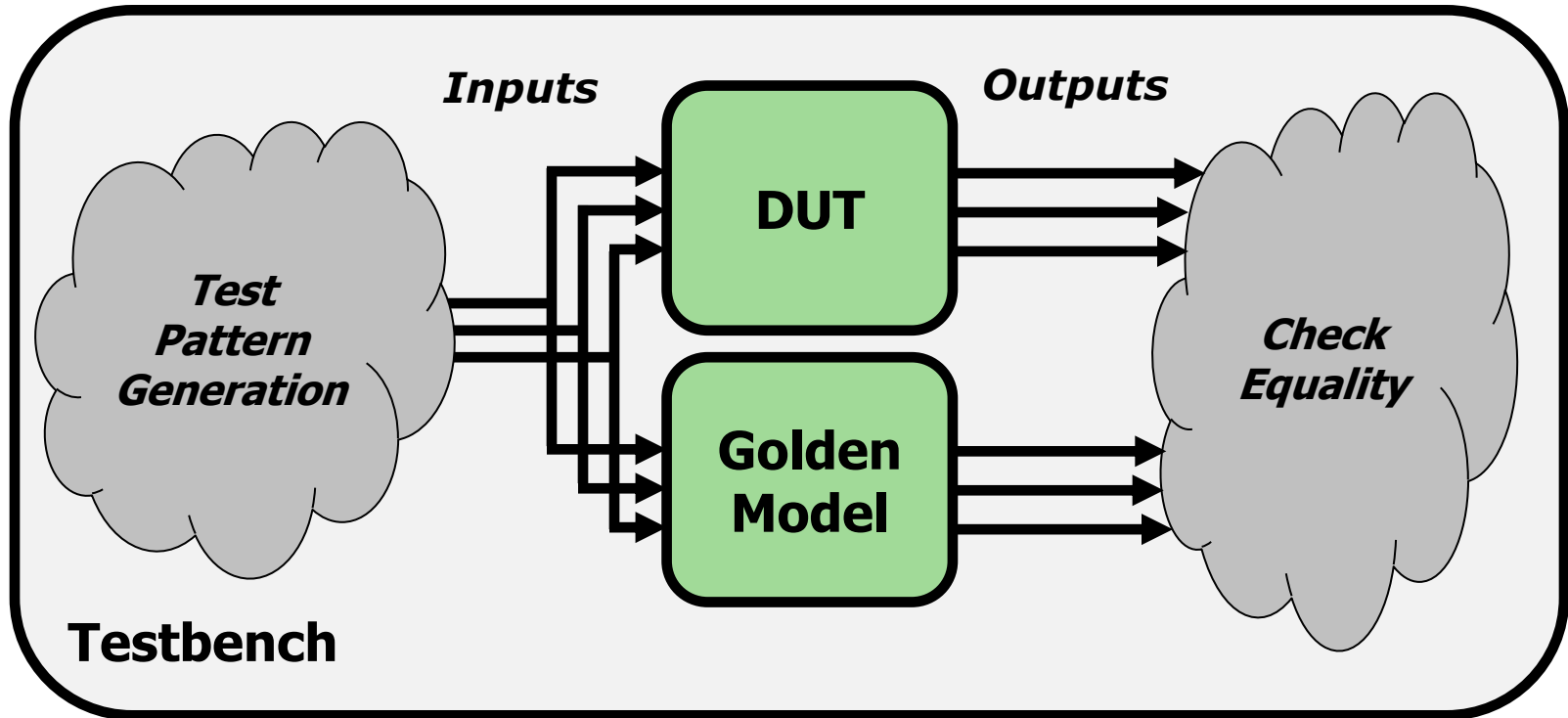
- A **golden model** represents the ideal circuit behavior
 - ❑ Must be developed, and might be **difficult** to write
 - ❑ Can be done in C, Perl, Python, Matlab or even in Verilog
- For our example circuit:

```
module golden_model(input  a, b, c,  
                    output y);  
    assign y = ~b & ~c | a & ~b; // high-level abstraction  
endmodule
```

- **Simpler** than our earlier gate-level description
 - ❑ Golden model is usually **easier to design and understand**
 - ❑ Golden model is much **easier to verify**

Automatic Testbench

- The DUT **output** is compared against the **golden model**



- **Challenge:** need to **generate inputs** to the designs
 - ❑ Sequential values to cover the entire input space?
 - ❑ Random values?

Automatic Testbench: Code

```
module testbench1();  
    ... // variable declarations, clock, etc.  
  
    // instantiate device under test  
    sillyfunction dut (a, b, c, y_dut);  
    golden_model gold (a, b, c, y_gold);  
  
    // instantiate test pattern generator  
    test_pattern_generator tgen (a, b, c, clk);  
  
    // check if y_dut is ever not equal to y_gold  
    always @(negedge clk)  
    begin  
        if(y_dut !== y_gold)  
            $display(...)  
    end  
endmodule
```


Automatic Testbench

■ Pros:

- ❑ Output checking is **fully automated**
- ❑ Could even compare **timing** using a **golden timing model**
- ❑ **Highly scalable** to as much simulation time as is feasible
 - Leads to **high coverage** of the input space
- ❑ Better **separation of roles**
 - Separate designers can work on the DUT and the golden model
 - DUT testing engineer can focus on **important test cases** instead of output checking

■ Cons:

- ❑ Creating a correct golden model may be (very) **difficult**
- ❑ Coming up with **good testing inputs** may be **difficult**

However, Even with Automatic Testing...

- How long would it take to test a **32-bit adder**?
 - In such an adder there are **64** inputs = 2^{64} possible inputs
 - If you test **one input in 1ns**, you can test 10^9 inputs per second
 - or 8.64×10^{14} inputs per day
 - or 3.15×10^{17} inputs per year
 - we would still need **58.5 years** to test all possibilities
- Brute force testing is **not feasible** for most circuits!
 - Need to prune the overall testing space
 - E.g., formal verification methods, choosing 'important cases'
- **Verification is a hard problem**

Part 5:

Timing Verification

Timing Verification Approaches

- High-level simulation (e.g., C, Verilog)
 - Can **model timing** using “#x” statements in the DUT
 - Useful for hierarchical modeling
 - Insert delays in FF’s, basic gates, memories, etc.
 - High level design will have some notion of timing
 - Usually **not as accurate** as real circuit timing
- Circuit-level timing verification
 - Need to first **synthesize** your design to actual circuits
 - No one general approach- very **design flow specific**
 - Your FPGA/ASIC/etc. technology has **special tool(s)** for this
 - E.g., Xilinx Vivado (what you’re using in labs)
 - E.g., Synopsys/Cadence Tools (for more advanced designs)

The Good News

- Tools will try to meet timing for you!
 - Setup times, hold times
 - Clock skews
 - ...
- They usually provide a '**timing report**' or '**timing summary**'
 - **Worst-case** delay paths
 - Maximum operation **frequency**
 - Any timing **errors** that were found

The Bad News

- The **tool can fail** to find a solution
 - Desired clock frequency is too **aggressive**
 - Can result in **setup time violation** on a particularly long path
 - **Too much logic** on clock paths
 - Introduces excessive **clock skew**
 - Timing issues with asynchronous logic
- The tool will provide (hopefully) **helpful errors**
 - Reports will contain paths that failed to meet timing
 - Gives a place from where to start debugging
- **Q:** How can we **fix timing errors**?

Meeting Timing Constraints

- Unfortunately, this is often a **manual, iterative** process
 - Meeting strict timing constraints (e.g., high performance designs) can be **tedious**
- Can try **synthesis/place-and-route** with different options
 - Different **random seeds**
 - Manually provided **hints** for place-and-route
- Can **manually optimize** the reported **problem paths**
 - Simplify **complicated logic**
 - Split up **long combinational logic paths**
 - Recall: fix hold time violations by adding **more** logic!

Meeting Timing Constraints: Principles

- Let's go back to the fundamentals
- Clock cycle time is determined by the maximum logic delay we can accommodate without violating timing constraints
- Good design principles
 - **Critical path design:** Minimize the maximum logic delay
→ Maximizes performance
 - **Balanced design:** Balance maximum logic delays across different parts of a system (i.e., between different pairs of flip flops)
→ No bottlenecks + minimizes wasted time
 - **Bread and butter design:** Optimize for the common case, but make sure non-common-cases do not overwhelm the design
→ Maximizes performance for realistic cases

Lecture Summary

- **Timing** in **combinational circuits**
 - ❑ Propagation delay and contamination delay
 - ❑ Glitches
- **Timing** in **sequential circuits**
 - ❑ Setup time and hold time
 - ❑ Determining how fast a circuit can operate
- **Circuit Verification**
 - ❑ How to make sure a circuit works correctly
 - ❑ Functional verification
 - ❑ Timing verification

Digital Design & Computer Arch.

Lecture 6: Timing and Verification

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Spring 2026

6 March 2026